

	Inspection Report	Doc.	DC-IR-260408
	Pot components	Rev.	0

Tran Hung Dao PROJECT

CONTRACTOR : NAM TIEN CONSTRUCTION
CONSULTING AND TRADING JSC
SUBCONTRACTOR : DAECHANG CORP.
PO NO : DAE-DC-202602-1

DAECHANG CORP.


 Jeongseong Kim **CEO**



0	2026.04.08	 H.Y. Gong	 S.H. Kang	ISSUE
REV	DATE	WRITTEN	APPROVED	DESCRIPTION

	Inspection Report	Doc.	DC-IR-260408
	Pot components	Rev.	0

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DAECHANG CORP.



Jeongseong Kim

 CEO

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	Pot components	Rev.	0

Dimension Inspection Report

DAECHANG CORP.



Jeongseong Kim

CEO

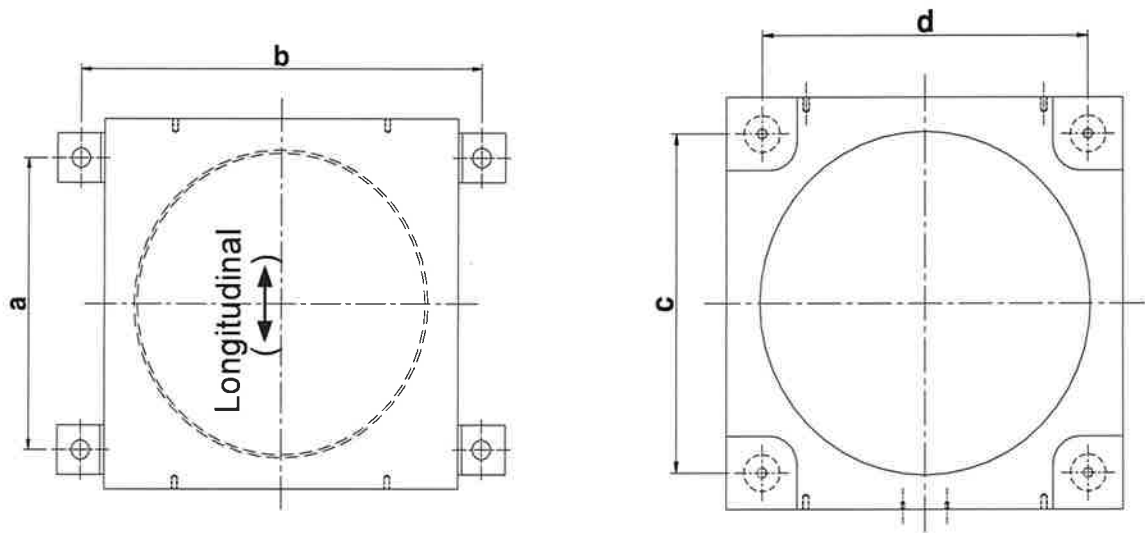
Dimension&Visual Inspection Report For Finished Product



완제품 치수검사 성적서

Dimension&Visual Inspection Report For Finished Product

보고서 번호 Report No.	26-012-F-001	날 짜 Date	2026. 4. 27 ~ 28
고 객 Client	NAM TIEN CONSTRUCTION CONSULTING AND TRADING JSC	공 사 명 Project Name	Tran Hung Dao Bridge
제 품 명 Item	9938/12986/994/1299/±0	도 면 Drawing Title.	Pot Bearing - Fixed Type
수 량 Quantity	2	검 사 기 준 Standard	AASHTO
비 고 Note	CD-10000	로 트 번 호 Lot No.	26-012



POINT	a	b	c	d														결 과
SPEC	710	1020	740	740														Result
Tolerance(mm)	±5	±5	±5	±5														
1	0	0	0	0														O.K.
2	0	0	0	0														O.K.

DAECHANG CORP.

Comment

SIGNATURE Jeongseong Kim CEC

You-Seok Kang.
DAECHANG CORP.
Q.A/Q.C Team_Staff

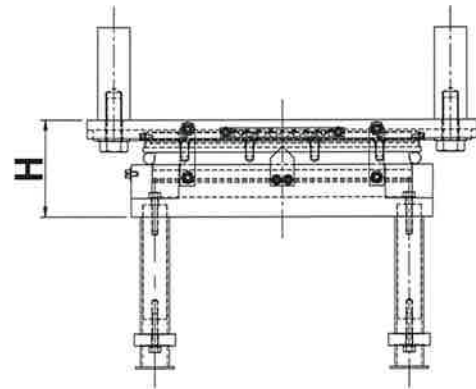
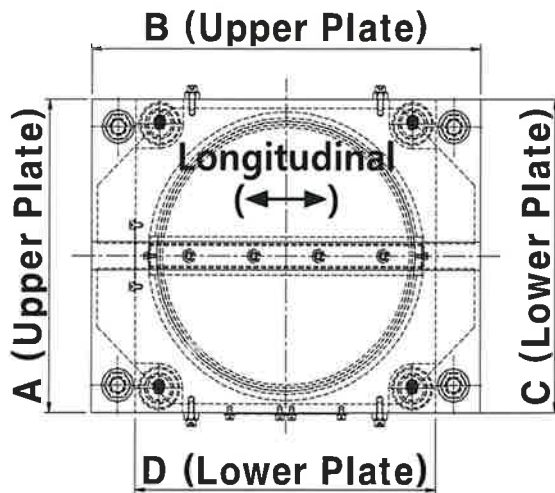
Jae-Hwan Lee.
DAECHANGCORP.
Q.A/Q.C Team_Manager

완제품 치수검사 성적서

Dimension&Visual Inspection Report For Finished Product

보고서 번호 Report No.	26-012-F-002	날 짜 Date	2026. 4. 27 ~ 28
고 객 Client	NAM TIEN CONSTRUCTION CONSULTING AND TRADING JSC	공 사 명 Project Name	Tran Hung Dao Bridge
제 품 명 Item	3584/5933/358/593/±51	도 면 Drawing Title.	Pot Bearing - Guide Type
수 량 Quantity	1	검 사 기 준 Standard	AASHTO
비 고 Note	DH-4000	로 트 번 호 Lot No.	26-012

VISUAL INSPECTION	
Welding	OK
Grinding	OK
Degree of Straight	OK
Degree of Verticality	OK



POINT	A	B	C	D	H						Film	결 과 Result
SPEC	510	660	510	510	157						MIN.	
Tolerance(mm)	0~3	0~3	0~3	0~3	0~6						250 μ m	Result
1	+1	+1	+1	+1	+2						336	O.K.
							</					

SIGNATURE

You-Seok Kang.
DAECHANG CORP.
Q.A/Q.C Team_Staff



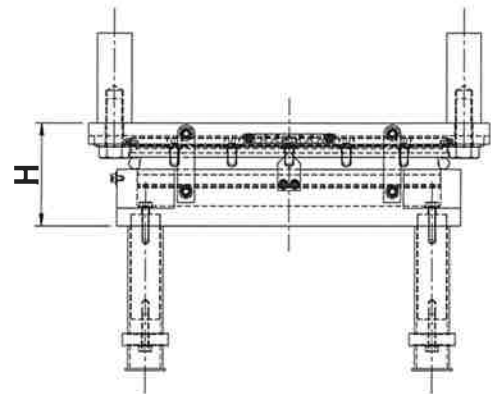
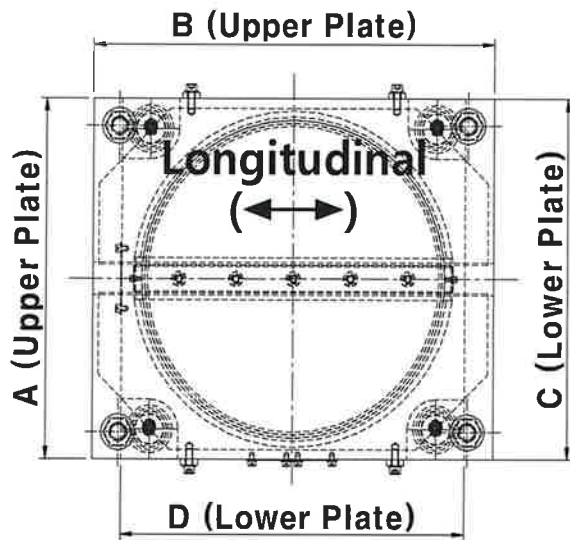
Jae-Hwan Lee.
DAECHANGCORP.
Q.A/Q.C Team_Manager



완제품 치수검사 성적서

Dimension&Visual Inspection Report For Finished Product

보고서 번호 Report No.	26-012-F-003	날 짜 Date	2026. 4. 27 ~ 28
고 객 Client	NAM TIEN CONSTRUCTION CONSULTING AND TRADING JSC	공 사 명 Project Name	Tran Hung Dao Bridge
제 품 명 Item	4981/6671/498/667/±20	도 면 Drawing Title.	Pot Bearing - Guide Type
수 량 Quantity	4	검 사 기 준 Standard	AASHTO
비 고 Note	DH-5000	로 트 번 호 Lot No.	26-012



VISUAL INSPECTION	
Welding	OK
Grinding	OK
Degree of Straight	OK
Degree of Verticality	OK

POINT	A	B	C	D	H							Film	결 과
SPEC	600	700	600	600	172							MIN.	Result
Tolerance(mm)	0~3	0~3	0~3	0~3	0~6							250/171	
1	+1	+1	+1	0	+2							361	O.K.
2	+1	+1	+1	+1	+1							295	O.K.
3	+1	+1	+1	+1	+2							368	O.K.
4	+1	+1	+1	+1	+2							312	O.K.
Comment													

DAECHANG CORP.

Jeongseong Kim CEO

SIGNATURE

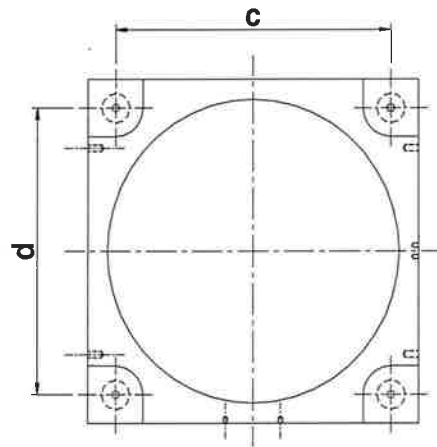
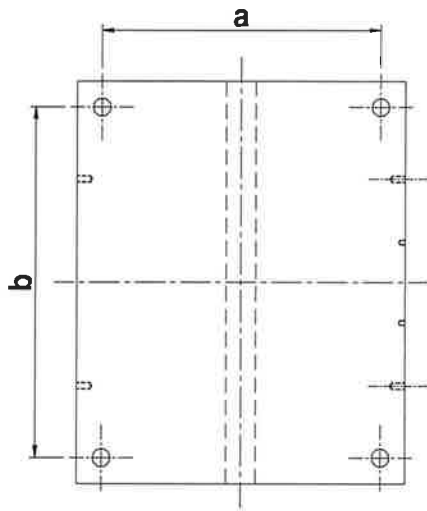
You-Seok Kang.
DAECHANG CORP.
Q.A/Q.C Team_Staff

Jae-Hwan Lee.
DAECHANGCORP.
Q.A/Q.C Team_Manager

완제품 치수검사 성적서

Dimension&Visual Inspection Report For Finished Product

보고서 번호 Report No.	26-012-F-003	날 짜 Date	2026. 4. 27 ~ 28
고 객 Client	NAM TIEN CONSTRUCTION CONSULTING AND TRADING JSC	공 사 명 Project Name	Tran Hung Dao Bridge
제 품 명 Item	4981/6671/498/667/±20	도 면 Drawing Title.	Pot Bearing - Guide Type
수 량 Quantity	4	검 사 기 준 Standard	AASHTO
비 고 Note	DH-5000	로 트 번 호 Lot No.	26-012



POINT	a	b	c	d									결 과
SPEC	510	610	500	500									Result
Tolerance(mm)	±5	±5	±5	±5									
1	0	0	0	0									O.K.
2	0	0	0	0									O.K.
3	0	0	0	0									O.K.
4	0	0	0	0									O.K.
Comment													

DAECHANG CORP.

SIGNATURE

Jeongseong Kim

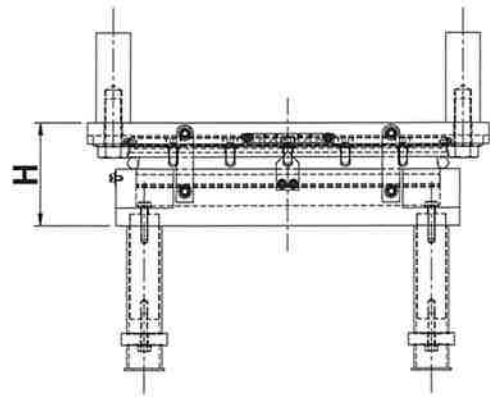
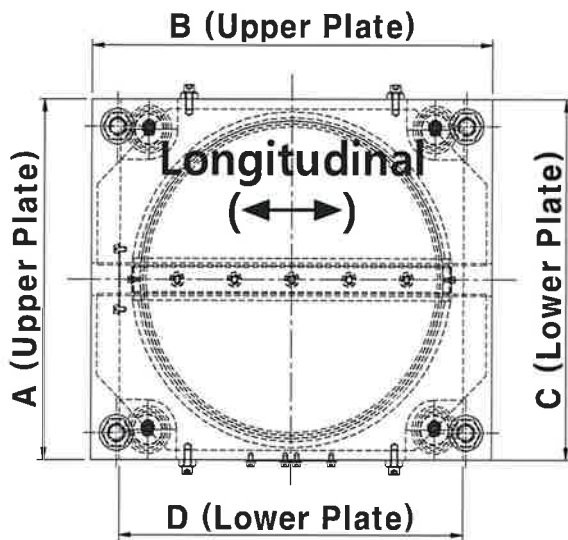
CEO

You-Seok Kang.
DAECHANG CORP.
Q.A/Q.C Team_Staff

Jae-Hwan Lee.
DAECHANGCORP.
Q.A/Q.C Team_Manager

Dimension&Visual Inspection Report For Finished Product

VISUAL INSPECTION	
Welding	OK
Grinding	OK
Degree of Straight	OK
Degree of Verticality	OK



POINT	A	B	C	D	H							Film	결 과 Result
SPEC	600	700	600	600	172							MIN.	
Tolerance(mm)	0~3	0~3	0~3	0~3	0~6							250 μ m	
1	+1	+1	+1	+1	+2							326	O.K.
2	+1	+1	+1	0	+2							354	O.K.
Comment													

DAECHANG CORP.

SIGNATURE

Jeongseong Kim

CEO

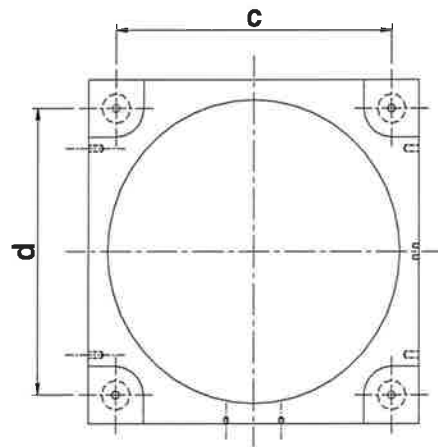
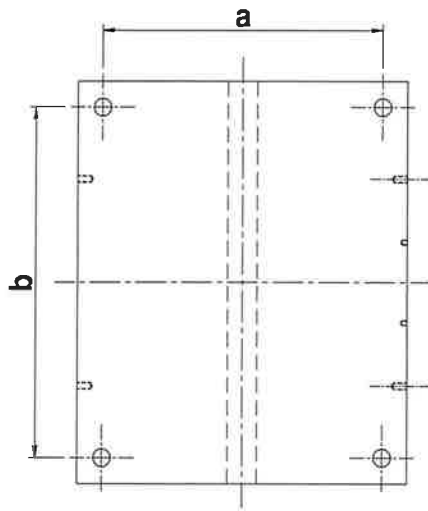
You-Seok Kang.
DAECHANG CORP.
Q.A/Q.C Team Staff

Jae-Hwan Lee.
DAECHANGCORP.
Q.A/Q.C Team Manager

완제품 치수검사 성적서

Dimension&Visual Inspection Report For Finished Product

보고서 번호 Report No.	26-012-F-004	날 짜 Date	2026. 4. 27 ~ 28
고 객 Client	NAM TIEN CONSTRUCTION CONSULTING AND TRADING JSC	공 사 명 Project Name	Tran Hung Dao Bridge
제 품 명 Item	5240/6926/524/693/±21	도 면 Drawing Title.	Pot Bearing - Guide Type
수 량 Quantity	2	검 사 기 준 Standard	AASHTO
비 고 Note	DH-6000	로 트 번 호 Lot No.	26-012



POINT	a	b	c	d													결 과
SPEC	510	610	500	500													Result
Tolerance(mm)	±5	±5	±5	±5													
1	0	0	0	0													O.K.
2	0	0	0	0													O.K.

DAECHANG CORP.

Jeongseong Kim CEO

Comment

SIGNATURE

You-Seok Kang.
DAECHANG CORP.
Q.A/Q.C Team_Staff

Jae-Hwan Lee.
DAECHANGCORP.
Q.A/Q.C Team_Manager

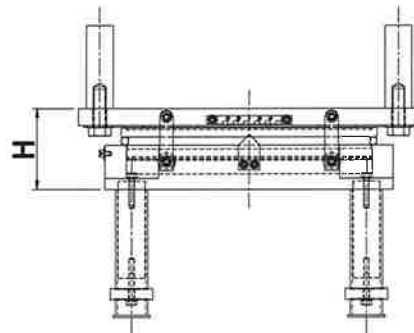
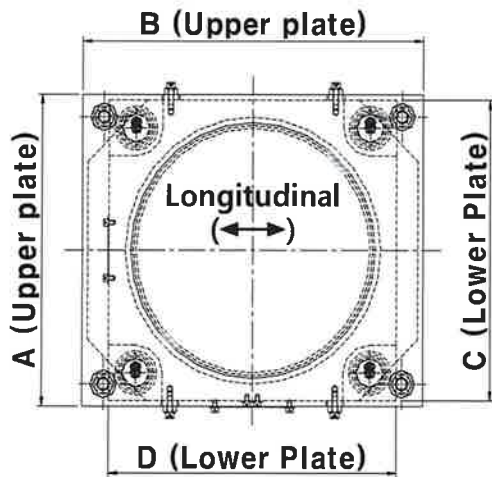
완제품 치수검사 성적서

Dimension&Visual Inspection Report For Finished Product

보고서 번호 Report No.	26-012-F-005	날 짜 Date	2026. 4. 27 ~ 28
고 객 Client	NAM TIEN CONSTRUCTION CONSULTING AND TRADING JSC	공 사 명 Project Name	Tran Hung Dao Bridge
제 품 명 Item	3986/4832/-/±10/±10	도 면 Drawing Title.	Pot Bearing - Multi Type
수 량 Quantity	2	검 사 기 준 Standard	AASHTO
비 고 Note	SH-4000	로 트 번 호 Lot No.	26-012

VISUAL INSPECTION

Welding	OK
Grinding	OK
Degree of Straight	OK
Degree of Verticality	OK



POINT	A	B	C	D	H						Film	결 과
SPEC	560	640	540	540	146						MIN.	
Tolerance(mm)	0~3	0~3	0~3	0~3	0~6						250μm	Result
1	+1	+1	+1	+1	+3						340	O.K.
2	+1	+1	+1	+1	+3						416	O.K.

DAECHANG CORP.

Jeongseong Kim

CEO

SIGNATURE

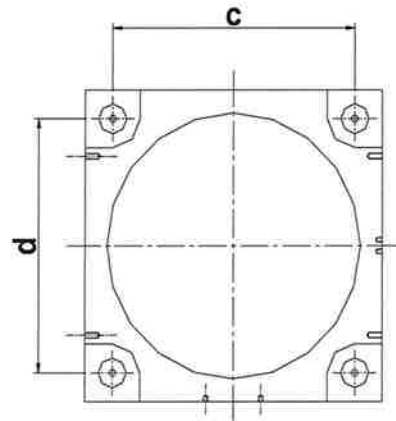
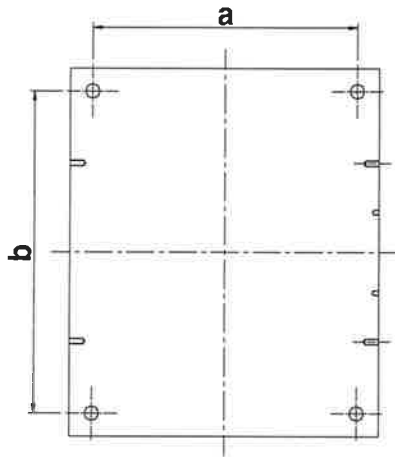
You-Seok Kang.
DAECHANG CORP.
Q.A/Q.C Team_Staff

Jae-Hwan Lee.
DAECHANGCORP.
Q.A/Q.C Team_Manager

완제품 치수검사 성적서

Dimension&Visual Inspection Report For Finished Product

보고서 번호 Report No.	26-012-F-005	날 짜 Date	2026. 4. 27 ~ 28
고 객 Client	NAM TIEN CONSTRUCTION CONSULTING AND TRADING JSC	공 사 명 Project Name	Tran Hung Dao Bridge
제 품 명 Item	3986/4832/-/±10/±10	도 면 Drawing Title.	Pot Bearing - Multi Type
수 량 Quantity	2	검 사 기 준 Standard	AASHTO
비 고 Note	SH-4000	로 트 번 호 Lot No.	26-012



POINT	a	b	c	d														결 과
SPEC	480	560	440	440														Result
Tolerance(mm)	±5	±5	±5	±5														
1	0	0	0	0														O.K.
2	0	0	0	0														O.K.
Comment																		

DAECHANG CORP.

Jeongseong Kim

CEO

SIGNATURE

You-Seok Kang.
DAECHANG CORP.
Q.A/Q.C Team_Staff

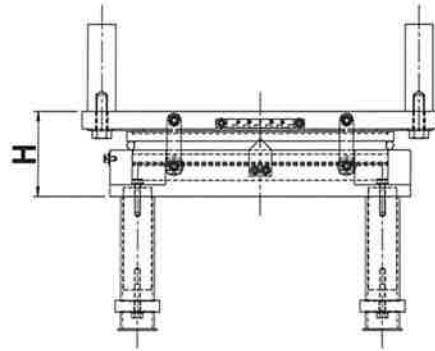
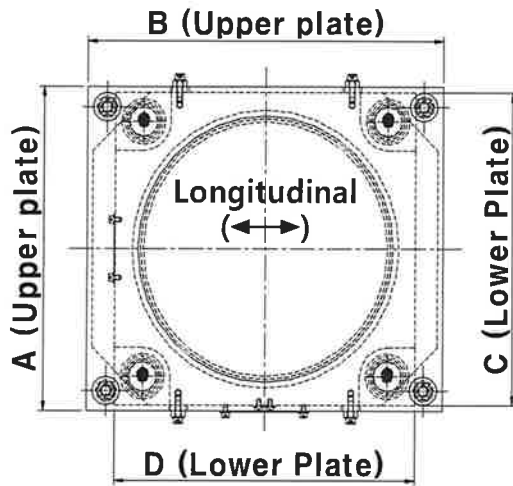
Jae-Hwan Lee.
DAECHANGCORP.
Q.A/Q.C Team_Manager

완제품 치수검사 성적서

Dimension&Visual Inspection Report For Finished Product

보고서 번호 Report No.	26-012-F-006	날 짜 Date	2026. 4. 27 ~ 28
고 객 Client	NAM TIEN CONSTRUCTION CONSULTING AND TRADING JSC	공 사 명 Project Name	Tran Hung Dao Bridge
제 품 명 Item	3198/7528/-/±48/±5	도 면 Drawing Title	Pot Bearing - Multi Type
수 량 Quantity	2	검 사 기 준 Standard	AASHTO
비 고 Note	SH-4000	로 트 번 호 Lot No.	26-012

VISUAL INSPECTION	
Welding	OK
Grinding	OK
Degree of Straight	OK
Degree of Verticality	OK



POINT	A	B	C	D	H							Film	결 과
SPEC	560	640	540	540	146							MIN.	Result
Tolerance(mm)	0~3	0~3	0~3	0~3	0~6							250μm	
1	+1	+1	+1	+1	+2							325	O.K.
2	+1	+1	+1	+1	+3							382	O.K.

Comment

SIGNATURE

Jeongseong Kim

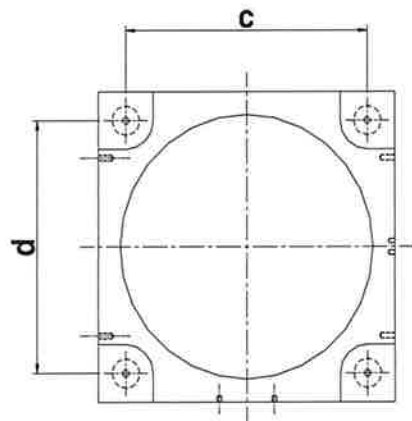
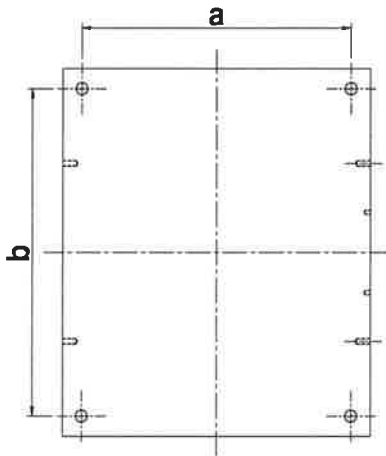
CEO

You-Seok Kang.
DAECHANG CORP.
Q.A/Q.C Team_Staff

Jae-Hwan Lee.
DAECHANGCORP.
Q.A/Q.C Team_Manager

Dimension&Visual Inspection Report For Finished Product

보고서 번호 Report No.	26-012-F-006	날 짜 Date	2026 . 4 . 27 ~ 28
고 객 Client	NAM TIEN CONSTRUCTION CONSULTING AND TRADING JSC	공 사 명 Project Name	Tran Hung Dao Bridge
제 품 명 Item	3198/7528/-/±48/±5	도 면 Drawing Title.	Pot Bearing - Multi Type
수 량 Quantity	2	검 사 기 준 Standard	AASHTO
비 고 Note	SH-4000	로 트 번 호 Lot No.	26-012



POINT	a	b	c	d								결 과 Result
SPEC	490	570	440	440								
Tolerance(mm)	±5	±5	±5	±5								
1	0	0	0	0								O.K.
2	0	0	0	0								O.K.

WATCO INC CORP

SIGNATURE

Jeongseong Kim

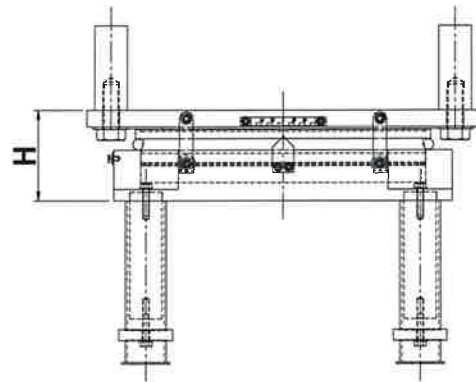
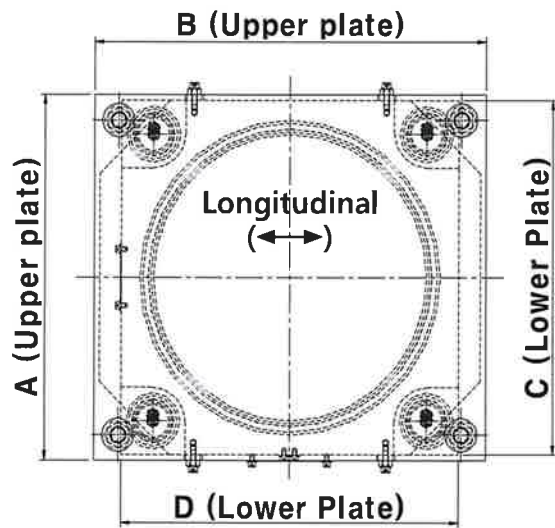
CEO

You-Seok Kang.
DAECHANG CORP.
Q.A/Q.C Team Staff

Jae-Hwan Lee.
DAECHANGCORP.
Q.A/Q.C Team_Manager

Dimension&Visual Inspection Report For Finished Product

VISUAL INSPECTION	
Welding	OK
Grinding	OK
Degree of Straight	OK
Degree of Verticality	OK



POINT	A	B	C	D	H							Film	결 과 Result
SPEC	650	730	630	630	161							MIN.	
Tolerance(mm)	0~3	0~3	0~3	0~3	0~6							250 μ m	
1	+1	+1	+1	+1	+3							349	O.K.
2	+1	+1	+1	+1	+2							285	O.K.
3	+1	+1	+1	+1	+2							375	O.K.
4	+1	+1	+1	+1	+2							362	O.K.
Comment													

SIGNATURE

Jeongseong Kim

CEO

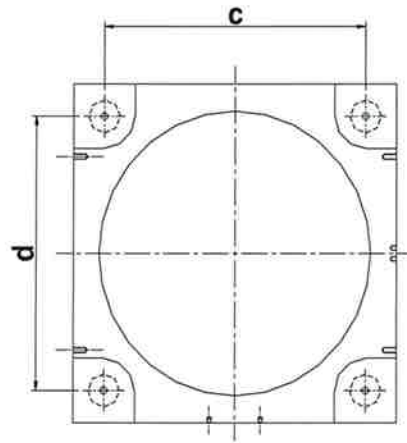
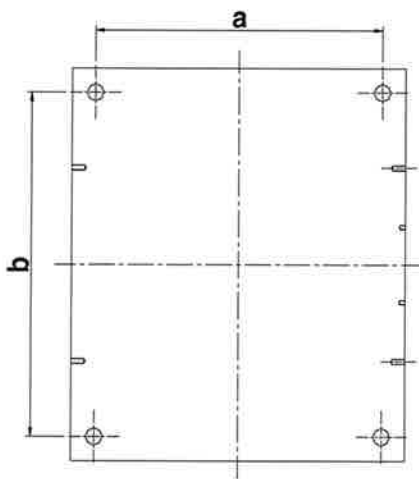
You-Seok Kang.
DAECHANG CORP.
Q.A/Q.C Team Staff

Jae-Hwan Lee.
DAECHANGCORP.
Q.A/Q.C Team Manager

완제품 치수검사 성적서

Dimension&Visual Inspection Report For Finished Product

보고서 번호 Report No.	26-012-F-007	날 짜 Date	2026 . 4 . 27 ~ 28
고 객 Client	NAM TIEN CONSTRUCTION CONSULTING AND TRADING JSC	공 사 명 Project Name	Tran Hung Dao Bridge
제 품 명 Item	5160/6974/-/±21/±10	도 면 Drawing Title.	Pot Bearing - Multi Type
수 량 Quantity	4	검 사 기 준 Standard	AASHTO
비 고 Note	SH-5000	로 트 번 호 Lot No.	26-012



POINT	a	b	c	d													결 과
SPEC	560	640	510	510													Result
Tolerance(mm)	±5	±5	±5	±5													
1	0	0	0	0													O.K.
2	0	0	0	0													O.K.
3	0	0	0	0													O.K.
4	0	0	0	0													O.K.
Comment																	



SIGNATURE																	Jeongseong Kim	CEO
You-Seok Kang.																	Jae-Hwan Lee.	
DAECHANG CORP.																	DAECHANGCORP.	
Q.A/Q.C Team_Staff																	Q.A/Q.C Team_Manager	

	Inspection Report	Doc.	DC-IR-260428
	Pot Bearing	Rev.	0

Materials Test Report

DAECHANG CORP.


 Jeongseong Kim CEO

SGS Korea Co., Ltd.

Issue No. : G-26-0157-R0

Construction Materials Testing Center

Random No. : SGSGFOBQ5Y

Address : 20,Dongtansandan 10-gil, and 12, Dongtan-myeon, Hwaseong-si, Gyeonggi-do, Korea, 18487

t +82(0)31 669-0600 Fax +82(0)31 646 0888

https://www.sgs.com/ko-kr

1. Applicant

◦Company : Daechang Corp

◦Address : 82, Soseok-ro, Daeso-myeon, Eumseong-gun, Chungcheongbuk-do, Korea

◦Receipt Date : November 06, 2025

2. Test Purpose : Maintenance of quality

3. Sample Description : Rubber Pad

4. Date of Test : November 20, 2025 ~ January 09, 2026

5. Location of Test : ☒ Permanent Testing Lab ☐ On Site Testing

(Address: 20, Dongtansandan 10-gil and 12, Dongtansandan 10-gil, Hwaseong-si, Gyeonggi-do)

6. Test Results :

No.	Test Items	Units	Test Methods	Results
1	Hardness (Type A, HS)	-	KS F 4424 : 1996	52
2	Tensile Strength	MPa		17.4
3	Elongation	%		520
4	Accelerated Air Oven Aging ((70 ± 1) °C, 168 h)	Change of Hardness (Type A, HS)		2
		Change of Tensile Strength		-9
		Change of Elongation		-8
5	Compression Set ((70 ± 1) °C, 22 h)	%		13
6	Ozone Resistance (500 MPa, (40 ± 2) °C, 20 % elongation, 100h)	-		No cracks

DAECHANG CORP.

Tester : Moon,SeungKwon

Tel : 031-646-0811

문승권

Technical Manager Jeongseong Kim

Tel : 031-646-0805

Jeongseong Kim CEO

- Remarks :
1. This test report is not related to KS Q ISO/IEC 17025 and KOLAS accreditation scope.
 2. The results shown in this report refer only to the sample(s) submitted by applicant. This test report can not be reproduced or used other than the test purpose stated in above.
 3. The authenticity of this test report can be verified through the SGS official website (<https://e-checking.kr.sgs.com/NewEchecking?pane=Construction>) or alternatively by scanning the QR code provided below.

Duplicate



Test Report Authentication
QR code

January 12, 2025

SGS Korea Co., Ltd.

Construction Materials Testing Center



Page 1 of 1

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CQP-708-001-F09 (03)_2025.10.13.

SGS Korea Co., Ltd.

20, Dongtansandan 10-gil and 12, Dongtansandan 10-gil, Hwaseong-si, Gyeonggi-do, Korea, 18487 t +82(0)31 646 0800 f +82(0)31 646 0888 www.sgs.com

Member of SGS Group(Société Générale de Surveillance)



INSPECTION CERTIFICATE

1. Manufacturer : DAECHANG Corp
2. Date of Manufacture : Feb. 23. 2026 ~ Present
3. Specimen : Rubber Pad
4. Date of Testing : Apr. 18. 2026 ~ Apr. 27. 2026
5. Test : Material Test - Rubber Pad
6. Project : Tran Hung Dao Project
7. Testing Location (Address) : 82, Soseok-ro, Daeso-myeon, Eumseong-gun,
Chungcheongbuk-do, Korea

8. Test Results

No.	Test		Unit	Req.	Test Spec.	Result
1	Tensile Strength		Mpa	≥ 15	KS F 4424 : 1996	18.1
2	Elongation		%	≥ 450		522
3	Hardness (HS)		-	46-55		51
4	Aging (70°C, 168h)	T.S Change	%	$\leq +15$		-7
5		El. Change	%	$\leq +40$		-9
6		Hardness Change	-	$\leq +15$		+23
7	Compression Set ((70±1)°C, 22h)		%	≤ 35		18
8	Ozone resistance ((500 MPa x (40 ± 2)°C x 100h))		-	No cracks	Vietnam Site Spec.	No cracks

9. Remarks

The test results shown in this report refer only to the sample(s).
This test report cannot be reproduced or used other than the test prupose stated in above.

Apr. 27. 2026



Rev.0

DAECHANG Corp

DAECHANG CORP. (4/20X297)

Jeongseong Kim
CEO

SGS Korea Co., Ltd.

Construction Materials Testing Center

Issue No. : G-26-0162-R0

Random No. : SGSnXxxKtP

Address : 20,Dongtansandan 10-gil, and 12, Dongtan-myeon, Hwaseong-si, Gyeonggi-do, Korea, 18487

t +82(0)31 669-0600 Fax +82(0)31 646 0888

https://www.sgs.com/ko-kr

1. Applicant

◦Company : Daechang Corp

◦Address : 82, Soseok-ro, Daeso-myeon, Eumseong-gun, Chungcheongbuk-do, Korea

◦Receipt Date : November 06, 2025

2. Test Purpose : Maintenance of quality

3. Sample Description : PTFE

4. Date of Test : November 20, 2025 ~ January 09, 2026

5. Location of Test : ☒ Permanent Testing Lab ☐ On Site Testing

(Address: 20, Dongtansandan 10-gil and 12, Dongtansandan 10-gil, Hwaseong-si, Gyeonggi-do)

6. Test Results :

No.	Test Items	Units	Test Methods	Results
1	Tensile Strength	MPa	KS F 4424 : 1996	20.7
2	Elongation	%		222
3	Specific Gravity	-		2.14

- end -

Duplicate

DAECHANG CORP.

Jeongseong Kim

CEO

Tester : Moon,SeungKwon

Tel : 031-646-0811

문승권

Technical Manager : Eom, JooHan

Tel : 031-646-0805

이준한

- Remarks :
1. This test report is not related to KS Q ISO/IEC 17025 and KOLAS accreditation scope.
 2. The results shown in this report refer only to the sample(s) submitted by applicant. This test report can not be reproduced or used other than the test purpose stated in above.
 3. The authenticity of this test report can be verified through the SGS official website (<https://e-checking.kr.sgs.com/NewEchecking?pane=Construction>) or alternatively by scanning the QR code provided below.



Test Report Authentication
QR code

January 12, 2025

SGS Korea Co., Ltd

Construction Materials Testing Center



Page 1 of 1

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CQP-708-001-F09 (03)_2025.10.13.

SGS Korea Co.,Ltd.

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Member of SGS Group(Société Générale de Surveillance)



Mill Sheet

▣ 업체명 : 주)대창이엔지

성적서 No. : CSE-2602-01

발급일자 : 2026. 02. 20.

시 료 명 : C2600W

용 도 : 품질관리

시험결과

1. 기계적 성질

시험항목	단 위	기준값	측정값	시 험 방 법
인장강도	N/mm ²	395-460	450	KS B 0802 : 0203
연 신 율	%	5 이상	9	

2. 화학 성분

시료명	Cu	Pb	Fe	Zinc
기준값	68.5~71.5	0.05Max	0.05Max	Rem.
측정값	69.5	0.0144	0.001	Rem.

이하여백

본 성적서는 당사에서 납품한 제품의 원자재시험 결과입니다.

 씨에스이엔지 대표 권순진

DAECHANG CORP.


Jeongseong Kim

CEO

SGS Korea Co., Ltd.

Construction Materials Testing Center

Issue No. : G-26-0108-R0

Random No. : SGSFvkruC5

Address : 20,Dongtansandan 10-gil, and 12, Dongtan-myeon, Hwaseong-si, Gyeonggi-do, Korea, 18487

t +82(0)31 669-0600 Fax +82(0)31 646 0888

https://www.sgs.com/ko-kr

1. Applicant

◦Company : Daechang Corp

◦Address : 82 Soseok-ro, Daeso-myeon, Eumseong-gun, Chungcheongbuk-do, Korea

◦Receipt Date : November 06, 2025

2. Test Purpose : Quality Control

3. Sample Description : Brass

4. Date of Test : December 08, 2025 ~ December 10, 2025

5. Location of Test : ☒ Permanent Testing Lab ☐ On Site Testing

(Address: 20, Dongtansandan 10-gil and 12, Dongtansandan 10-gil, Hwaseong-si, Gyeonggi-do)

6. Test Results :

- Refer to the next page -

DAECHANG CORP.


Jeongseong Kim CEO

Tester : Yoo, Lia

Tel : 031-646-0810

Technical Manager : : Kang HyeokJun

Tel : 031-646-0823

Remarks :

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January 08, 2026

SGS Korea Co., Ltd
Construction Materials Testing Center



No.	Test Items		Units	Test Methods	Results
1	Tensile Strength		N/mm ²	KS B 0802 : 2003	450
2	Elongation		%	KS B 0802 : 2003	11
3	Chemical Composition	Cu	%	KS D 1651 : 2020	65.29
		Pb	%	KS D 1651 : 2020	0.048
		Fe	%	KS D 1651 : 2020	0.010
		Zn	%	KS D 1651 : 2020	34.40

- Tensile test specimen : No.9

- End-

DAECHANG CORP.


Jeongseong Kim  CEO



Test Report

SGS Korea Co., Ltd.

Issue No. : G-26-0163-R0

Construction Materials Testing Center

Random No. : SGSUBssxgZ

Address : 20, Dongtansandan 10-gil, and 12, Dongtan-myeon, Hwaseong-si, Gyeonggi-do, Korea, 18487

t +82(0)31 669-0600 Fax +82(0)31 646 0888

<https://www.sgs.com/ko-kr>

1. Applicant

◦Company : Daechang Corp

◦Address : 82, Soseok-ro, Daeso-myeon, Eumseong-gun, Chungcheongbuk-do, Korea

◦Receipt Date : November 06, 2025

2. Test Purpose : Maintenance of quality

3. Sample Description : Seal Ring(C08)

4. Date of Test : November 20, 2025 ~ January 09, 2026

5. Location of Test : ☒ Permanent Testing Lab ☐ On Site Testing

(Address: 20, Dongtansandan 10-gil and 12, Dongtansandan 10-gil, Hwaseong-si, Gyeonggi-do)

6. Test Results :

DAECHANG CORP.

Jeongseong Kim

CEO

No.	Test Items	Units	Test Methods	Results
1	Static Shear Modulus	MPa	KS M 6617 : 2021	0.85
2	Elongation	%		529
3	Oil Resistance (No.3 oil, (100 ± 1) °C, 70 h) Volume Change Rate	%		32
4	Aging Test ((100 ± 1) °C, 70 h)	Stress Change Rate at 25 % Elongation		15
		Elongation Change Rate		-12
5	Ozone Resistance ((40 ± 2) °C, 72 h, (50 ± 5) pphm, 20 %, Elongation)	-		No cracks
6	Compression Set ((100 ± 1) °C, 22 h)	%		15

- end -

Duplicate

Tester : Moon,SeungKwon

Tel : 031-646-0811

문승권

Technical Manager : Eom, JooHan

Tel : 031-646-0805

이준한

- Remarks :
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January 12, 2025

SGS Korea Co., Ltd

Construction Materials Testing Center



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 DAECHANG CORP. www.daechangcorp.com	Inspection Report	Doc.	DC-IR-260428
	Pot Bearing	Rev.	0

Mill Test Certificate

DAECHANG CORP.

Jeongseong Kim CEO

HYUNDAI
BNG

Inspection No 20241126160830

Specification
KS-D-3698

2022

Date of issue
NOV.26.2024

<http://www.bngsteel.com>

Changwon Works
124 Jeokhyeon-Ro, Seongsan-gu, Changwon,
Gyeongnam, Korea, 51707
Tel/fax : 055 - 268 - 4133



Total	5	86,350
-------	---	--------

DAECHANG CORP

1151 ¹⁰⁰W00501091

15

S.K YANG

Management
Systems
1504 EOB 031
1504 EOB 031

choreographed. Yes, we can! Let's

매년 10월 1일부터 12월 31일까지

We hereby certify that the Material described above has been tested and complies with the terms of the order contract.

HYUNDAI
BNG STEEL

Inspection No 20241017154701

Specification
KS-D-3698

STS304

28

124 Jeokhyeon-ro, Seongsan-gu, Changwon,
Gyeongnam, Korea, 51707
Telefax : 055 - 268 - 4133



124 Jeokhyeon-Ro, Seongsan-gu, Changwon,

Telefax: 055 - 268 - 4133

Telefax: 055 - 268 - 4133

Total	6	105,324
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DAECHANG CORP.

WILLIAMSON

CC

모든 증명서에 명시된 용도와 사용시 안전상의 문제가 발생 할 수 있으며, 저사 중영서 위, 보조시 사문서 위조(형법231조)로 파벌 이익을 당할 수 있습니다.

We hereby certify that the Material described above has been tested and complies with the terms of the order contract.



Management
System
ISO 9001:2015
ISO 14001:2015

S.K. YANG

Online Control Team Team Leader

검사증명서
MILL TEST CERTIFICATE



당진제철소 : 흥남 당진시 송악읍 북부산업로 1480
1480 Bukbusanup-ro, Songak-Eup, Dangjin-Si, Chungnam, Korea



주문번호
Order No. : D240415983

품명
Commodity : Plate

제품규격
Specification : KS D3515 SM355A

고객사
Customer : KASKO

주문자
Contractor : KASKO

증명서 번호
Certificate No. : 20240413-PN-0120-002

발행일자
Date of Issue : 2024-04-13

제품사수 Dimension	수량 Quantity	중량 Weight (Kg)	제강번호 Heat No.	제품번호 Product No.	Position	인장시험 Tensile Test						충격시험 Impact Test V-Notch +20.0 °C			화학성분 Chemical Composition (%)			단위 Unit					
						YP	TS	EL	YP-EL	PA	YR	YP	Energy	SF	LE	C	Si	Mn	P	S	CEQ	3 : x100	5 : x100,000
						(N/mm ²)	(%)	(%)	(%)	(%)	(%)	Mid	(Joule)	(%)	(mm)	4	3	3	4	4	3		
35.00 x 2440 x 3850	3	7,743	989368	FD10224503	T	362.7	530.3	29				U	A-203			L	1.664	289	1.310	118	22	396	
*** Sub Total (009) *** 3 7,743(Kg)																							
*** Condition of supply : As-rolled ***																							
*** Grade Total *** 3 7,743(Kg) Ceq(CEV): G = C + Mn/6 + (Ni + Cu)/15 + (Cr + Mo + V)/5																							
*** Last Item ***																							

DAECHANG CORP.
Jeongseong Kim
CEO

* Position - T:Top, M:Middle, B:Bottom
* Tensile test - Direction: Transverse, Gauge length: 200mm (Rectangular), YP Mid : 2 (0.2% off-set), S (0.5% underload), U (Upper yield point), L (Lower yield point)
* Impact test - Direction: Longitudinal, SF: Shear fracture, LE: Lateral Expansion
* Division - P: Product analysis, L: Ladle analysis

WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS

Signature
Chief Of Plate Quality Team

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* You are able to check an original electronic document by logging in to the customer portal of Hyundai-Steel. (http://sm.hyundai-steel.com/rs/cm/login.sp) * ORcode scanner App : 'ORcal'
* (사용상 주의사항) 본 검사증명서에 필기된 규격 외 사용시 제품손상 및 안전성 문제 발생 가능성이 있습니다.
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검사증명서
MILL TEST CERTIFICATE



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1480 Buktusanup-Ro, Songak-Eup, Dangjin-Si, Chungnam, Korea



주문번호
Order No. : D240520955

품명
Commodity : Plate
고객사
Customer : KASKO

제품규격
Specification : KS D3515 SM355A
주문자
Contractor : KASKO

증명서 번호
Certificate No. : 20240524-PN-0078

발행일자
Date of Issue : 2024-05-24

제품치수 Dimension	수량 Quantity	중량 Weight (Kg)	제강번호 Heat No.	제품번호 Product No.	Position	인장시험 Tensile Test						충격시험 Impact Test V-Notch +20.0 °C			화학성분 Chemical Composition(%)					단위 Unit					
						YP (N/mm ²) ()	TS (%)	EL (%)	YP-EL (%)	FA (%)	YR (%)	YP Mid	Energy (Joule)	SF (%)	LE (mm)	Division	C	Si	Mn	P	S	CEQ			
37.00 x 2438 x 6000	1	4,248	S96526	FC70056603	T	363.6	545.8	29				2	A: 248			L	1,609	292	1,259	118	33	382			
*** Sub Total(001) *** 1 4,248 (Kg)																									
*** Condition of supply : As-Filled ***																									
*** Grade Total *** 1 4,248 (Kg) Ceq(CEV): G = C + Mn/6 + (N + Cu)/15 + (Cr + Mo + V)/5																									
*** Last Item ***																									

DAECHANG CORP.

Jeongseong Kim CEO

NOTE	* Position - T: Top, M: Middle, B: Bottom
	* Tensile test - Direction: Transverse, Gauge length: 200mm (Rectangular), YP Mtd : 2 (0.2% off-set), 5 (0.5% underload), U (Upper yield point), L (Lower yield point)
	* Impact test - Direction: Longitudinal, SF: Shear fracture, LE: Lateral Expansion
	* Division - P: Product analysis, L: Ladle analysis

WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS	
Signature	
Chief Of Plate Quality Team	

* 본 검사증명서는 원본인 전자문서(전자서명 정보 포함)로부터 출력된 사본입니다. 전자문서의 내용은 원본과 동일하며, 고객포털에서 확인하실 수 있습니다. (http://sm.hyundai-steel.com/cs/cm/login.jsp)
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검사증명서 MILL TEST CERTIFICATE



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1480 Bukbuseonup-Ro, Songak-Eup, Dangjin-Si, Chungnam, Korea



주문번호
Order No. : D230421196

품명
Commodity : Plate 고객사
Customer : KASKO

제품규격
Specification : KS D3515 SM355A 주문자
Contractor : KASKO

증명서 번호
Certificate No. : 20230411-PN-0175-001

발행일자
Date of Issue : 2023-04-11

제품치수 Dimension	수량 Quantity	중량 Weight (Kg)	재질번호 Heat No.	제품번호 Product No.	Position	인장시험 Tensile Test						충격시험 Impact Test V-Notch +20.0 °C			화학성분 Chemical Composition(%)				단위 Unit											
						YP (N/mm ²) ()	TS (%)	EL (%)	Y _P - EL (%)	RA (%)	YR (%)	Y _P Mtd	Energy/ (Joule)	SF (%)	LE (mm)	C	S	Mn	P	S	CEQ	2 : x 100 3 : x 1,000 4 : x 10,000 5 : x 100,000	Division	L	1,550	351	1,497	131	34	417
50.00 x 2570 x 10820	1	10,915	S63420	FC59105801	T	308.9	564.1	33				U	A: 211																	
*** Sub Total(003) *** 1 10,915(Kg)																														
*** Condition of supply : As-Rolled ***																														
*** Grade Total *** 1 10,915(Kg) Ceq(CEV), G = C + Mn/6 + (Ni + Cu)/15 + (Cr + Mo + V)/5																														
*** Last Item ***																														

DAECHANG CORP.

230512

Jeongseong Kim CEO

- Position - T:Top, M:Middle, B:Bottom
Tensile test - Direction:Transverse, Gauge length:50mm (Round), YP Mtd : 2 (0.2% off-set), S (0.5% underload), U (Upper yield point), L (Lower yield point)
Impact test - Direction:Longitudinal, SF:Shear fracture, LE:Lateral Expansion
Division - P:Product analysis, L:Ladle analysis

Signature

Kim in

Chief Of Plate Quality Team

WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS
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당진제철소 : 중남 당진시 송악읍 북부산업로 1480
1480 Bulksanup-Ro, Songak-Eup, Dengjin-Si, Chungnam, Korea



주문번호
Order No.

: D240220262

품명
Commodity

: Plate

제품규격
Specification

: KS D3515 SM355A

고객사
Customer

: HANIL IRON & STEEL CO., LTD.

주문자
Contractor

: HANIL IRON & STEEL CO., LTD.

증명서 번호
Certificate No.

: 20240409-PN-0056-006

발행일자
Date of Issue

: 2024-04-09

제품치수 Dimension	수량 Quantity	중량 Weight (Kg)	재강번호 Heat No.	제품번호 Product No.	Position	인장시험 Tensile Test						충격시험 Impact Test V Notch +20.0 °C		화학성분 Chemical Composition(%)				단위 Unit																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
						YP			EL			YP-EL			RA		YR		YP		Energy (Joule)	SF (%)	LE (mm)	Division		C	Si	Mn	P	S	CEQ																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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*** Condition of supply : As-rolled ***

*** Grade Total *** 15 44,680(Kg) Ceq(CEI). G=C+Mn/6+(Ni+Cu)/15+(Cr+Mo+V)/5

*** Last item ***

DAECHANG CORP.



Jeongseong Kim CEO

- * Position - T:Top, M:Middle, B:Bottom
* Tensile test - Direction:Transverse, Gauge length:50mm (Round), YP Mid : 2 (0.2% off-set), 5 (0.5% underload), U (Upper yield point), L (Lower yield point)
* Impact test - Direction:Longitudinal, Size:10mmX10mm, SF:Shear fracture, LE:Lateral Expansion
* Division - P:Product analysis, L:Ladle analysis

* This Mill Test Certificate is issued in accordance with EN 10204 Type 3.1

WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS

Signature

Kim in

Chief Of Plate Quality Team

- * 본 검사증명서는 원본인 전자문서(전자서명 정보 포함)로부터 출력된 사본입니다. 전자문서의 내용은 위대체질 고객포도에서 확인하실 수 있습니다.(http://sm.hyundai-steel.com/cs/cm/login.jsp)
* This Mill Test Certificate is a copy that has been printed from original electronic document(with digital signing). You are able to check an original electronic document at the Hyundai Steel Customer portal (http://sm.hyundai-steel.com/cs/cm/login.jsp) * QRcode scanner App : 'QRreal'
* (사용상 주의사항) 본 검사증명서에 양기된 규격과 사용시 제품손상 및 안전상 문제가 발생할 수 있습니다.
* (Caution for Use) Damage on products and safety problems may arise if used other than to the spec stated in this Mill Test Certification.



검사증명서

MILL TEST CERTIFICATE



당진제철소 : 충남 당진시 송악읍 북부신인로 1480
1480 Bulksanup-Ro, Songak-Eup, Dangjin-Si, Chungnam, Korea

수량
Order No. : D231112260

품명
Commodity : Plate

제품규격
Specification : KS D3515 SM355A

고객사
Customer : KASKO

주문자
Contractor : KASKO

증명서 번호
Certificate No. : 20231121-PN-0059-001

발행일자
Date of Issue : 2023-11-21

제품치수 Dimension	수량 Quantity	중량 Weight (Kg)	제강번호 Heat No.	제품번호 Product No.	Position	인장시험 Tensile Test						충격시험 Impact Test V-Notch +20.0 °C				화학성분 Chemical Composition(%)		단위 Unit					
						YP	TS	EL	YP-EL	RA	YR	YP	SF	LE	Divison	C	Si	Mn	P	S	CEQ	2 : x 100	3 : x 1,000
60.00 x 2380 x 8440	1	9,462	573866	FC53812001	T	391.6	553.0	33				U	A: 201		L	1,606	372	1,467	153	23	418		
*** Sub Total(001) *** 1 9,462(Kg)																							
*** Condition of supply : As-rolled ***																							
*** Grade Total *** 1 9,462(Kg) Ceq(CEV), G = C + Mn/6 + (N + Cu)/15 + (Cr + Mo + V)/5																							
*** Last item ***																							

DAECHANG CORP.



Jeongseong Kim CEO

* Position - T:Top, M:Middle, B:Bottom
* Tensile test - Direction:Transverse, Gauge length:50mm (Round), YP Mtd : 2 (0.2% off-set), 5 (0.5% underload), U (Upper yield point), L (Lower yield point)
* Impact test - Direction:Longitudinal, SF:Shear fracture, LE:Lateral Expansion
* Division - P:Product analysis, L:Ladle analysis

WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS

Signature
Chief Of Plate Quality Team

상기의 제품은 검사의 결과 지정된 규격에 합격한 것을 증명합니다.



Mill Test Certificate/시험증서

Date of Issue/발행일자 : Apr., 12, 2024

Order No./계약번호 : 01S6295085

PO No./주문번호 : P*2307-114

Supplier : POSCO MOBILITY SOLUTION
/주문자 : CORPORATION PLATE GWANGYANG

Commodity : PLATE
/품명

Customer : POSCO MOBILITY SOLUTION
/고객사

Spec & Type : KS D3515 SM355A
/규격

Size/치수	Product No. /제품번호	Quantity /수량	Weight /중량 (kg)	Heat No. /제강번호	Country of Melt & Pour	Position	Tensile Test /인장시험			Impact Test /충격시험 /충격시합 V Notch +20 °C Energy (Joule)	Division	Chemical Composition/화학성분						
							YP (%)	TS (%)	EL (%)			C	SI	Mn	P	S	CEQ	PCM
05x2438x096	PP42038601	3	22,750	SU08412	KOR	T	359	559	34	1.184 2.180 3.115 A.180	L	0.1540	0.391	1.432	0.0126	0.0016	0.41	0.24
*** Sub Total (030) ***																		
*** Grade Total***																		
*** Grand Total***																		
=== Last Item ===																		

* Position - T : Top, M : Middle, B : Bottom
* Tensile Test Direction : 1/4T Transversal, Gauge Length : 50 mm(Round),
YP Method : Upper Point
* Impact Test Direction : 1/4T Longitudinal, Full-Size : 10 mmX10 mm
* Division - L : Ladle Analysis
* Supply Condition : As-Rolled unless otherwise Heat Treated.

This material has been fully killed and made by basic oxygen process. This material is fine grained steel.
This material has been made by vacuum degassing process.
Test Certificate is issued according to ISO 10474/EN 10204 3.1.
We, POSCO, hereby certify that the products specified in the present MTC have been fully processed in South Korea. (Facility and address information is stated below)
We hereby certify that the material herein has been made in accordance with the order and above specification.

본 검사증명서 위변조시 사문서 무효28항법 제231조29로 불이익을 당할 수 있으며, 본 증서는 KOLAS와 판권이 없습니다.

Surveyor To :

Kang, Daegu



DAECHANG CORP.
Jeongseong Kim
CEO

Order No./계약번호 : 01S6521430

Supplier : POSCO MOBILITY SOLUTION
/주분자 : CORPORATION PLATE GWANGYANG

Customer : POSCO MOBILITY SOLUTION
/고객사

PO No./주문번호 : P*2312-15

Commodity : PLATE
/품명

Spec & Type : KS D3515 SM355A
/규격

Size/치수		Product No. /제품번호	Quantity /수량	Weight /중량 (kg)	Heat No. /재간번호	Country of Melt & Pour	Position	Tensile Test /인장시험			Impact Test /충격시험	Division	Chemical Composition/화합 성분													
65x2438x6096		PB91108001	1	7,582	SB95680	KOR	T	YP 349	TS 525	EL 37	V Notch +20 °C Energy (Joule)	L	C (%)	0.1546	Si (%)	0.402	Mn (%)	1.422	P (%)	0.0090	S (%)	0.0028	CEQ (%)	0.40	PCM (%)	0.24
*** Sub Total (020) ***			1	7,582 (kg)									CEQ (G) = C + Mn/6 + (Ni + Cu)/15 + (Cr + Mo + V)/5 PCM (B) = C + Si/30 + (Mn + Cu + Cr)/20 + Ni/60 + Mo/15 + V/10 + 5*B													
*** Grade Total ***			1	7,582 (kg)																						

DAECHANG COR



Jeongseong Kim

DAECHANG CORP.
Jeongseong Kim
CEO

* Position - T : Top, M : Middle, B : Bottom
* Tensile Test, Direction : 1/4T Transversal, Gauge Length : 50 mm(Round),
YP Method : Upper Point
* Impact Test, Direction : 1/4T Longitudinal, Full-Size : 10 mmX10 mm
* Division - L : Ladle Analysis
* Supply Condition : As-Rolled unless otherwise Heat Treated.

This material has been fully killed and made by basic oxygen process. This material is fine grained steel.
This material has been made by vacuum degassing process.
Test Certificate is issued according to ISO 10474/EN 10204 3.1.
We, POSCO, hereby certify that the products specified in the present MTC have been fully processed in South Korea. (Facility and address information is stated below)
We hereby certify that the material herein has been made in accordance with the order and above specification.



본 검사증명서 인본조시 사본서 위조28항법 제231조20로 불이익을 당할 수 있으며, 본 증서는 KOLAS와 관련이 없습니다.

Surveyor To:

B.S. Kim

Order No./계약번호 : 01S6521430

Supplier /주분자 : POSCO MOBILITY SOLUTION
CORPORATION PLATE GWANGYANG

Customer /고객사 : POSCO MOBILITY SOLUTION

PO No./주문번호 : P*2312-I5

Commodity /품명 : PLATE

Spec & Type : KS D3515 SM355A
규격

Size/치수	Product No. 제품번호	Quantity 수량	Weight 중량 (kg)	Heat No. 재강번호	Country of Melt & Pour	Position	Tensile Test 인장시험 YP TS EL (MPa) (%)	Impact Test 충격시험 V Notch +20 °C Energy (Joule)	Division	Chemical Composition/화합성분 C Si Mn P S CEQ PCM (%)
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DAECHANG CORP.



Jeongseong Kim

CEO

* Position - T : Top, M : Middle, B : Bottom
* Tensile Test, Direction : 1/4T Transversal, Gauge Length : 50 mm(Round),
YP Method : Upper Point
* Impact Test, Direction : 1/4T Longitudinal, Full-Size : 10 mmX10 mm
* Division - L : Ladle Analysis
* Supply Condition : As-Rolled unless otherwise Heat Treated.

This material has been fully killed and made by basic oxygen process. This material is fine grained steel.
This material has been made by vacuum degassing process.
Test Certificate is issued according to ISO 10474/EN 10204 3.1.
We, POSCO, hereby certify that the products specified in the present MTC have been fully processed in South Korea. (Facility and address information is stated below)
We hereby certify that the material herein has been made in accordance with the order and above specification.

본 검사증명서 메탄조시 사본서 위조28형호 제231조20로 불이익을 당할 수 있으며, 본 증서는 KOLAS와 관련이 없습니다.



Surveyor To :

B.S. Kim

Order No./계약번호 : 01S6947231

PO No./주문번호 : 01S6947231

Supplier : eSteel4U
/주문자Commodity : PLATE
/품명Customer : ESTEEL4U
/고객사Spec & Type : KS-SM355A
/규격

Size/지수		Product No. 제품번호	Quantity 수량	Weight 중량 (kg)	Heat No. 제강번호	Country of Melt & Pour	Position	Tensile Test 인장시험 YP TS EL (MPa) (%)		Impact Test 충격시험 V Notch +20 °C Energy (Joule)	Division	Chemical Composition/화합성분 C Si Mn P S CEQ PCM (%)					
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DAECHANG CORP.

CEO

Jeongseong Kim

Position - T : Top, M : Middle, B : Bottom
Tensile Test. Direction : 1/4T Transversal, Gauge Length : 50 mm(Round),
YP Method : Upper Point
Impact Test. Direction : 1/4T Longitudinal, Full-Size : 10 mmX10 mm
Division - L : Ladle Analysis
Supply Condition : As-Rolled unless otherwise Heat Treated.

We guarantee particular outlined in the specification for inventory sales.
This material has been fully killed and made by basic oxygen process.
Test Certificate is issued according to ISO 10474/EN 10204 3.1.
We, POSCO, hereby certify that the products specified in the present MTC have been fully processed in South Korea. (Facility and address information is stated below)
We hereby certify that the material herein has been made in accordance with the order and above specification.

본 검사증서 위변조시 사문서 위조28형법 제231조29로 불이익을 당할 수 있으며, 본 증서는 KOLAS와 관련이 없습니다.

poscomtc@posco.com
+82-2-3457-0114

Surveyor To :

Kang, Deokgu

Order No./계약번호 : 01S5912290

Supplier : eSteel4U
/주문자

PO No./주문번호 : 01S5912290

Commodity : PLATE
/품명Customer : ESTEEL4U
/고객사Spec & Type : KS-SM355A
/규격

Size/치수	Product No. /제품번호	Quantity /수량	Weight /중량 (kg)	Heat No. /제강번호	Position	Tensile Test /인장시험		Impact Test /충격시험		Division	Chemical Composition/화합성분				
						YP (%)	TS (%)	EL (%)	V Notch +20 °C Energy (Joule)		C (%)	Si (%)	Mn (%)	P (%)	S (%)
140x2401x10202	PB75078101	1	26,920	SF60234	T	337	547	34	1: 169 2: 146 3: 174 A: 163	L	0.1651	0.454	1.551	0.0063	0.0030
*** Sub Total (010) ***		1	26,920 (kg)												
*** Grade Total ***		1	26,920 (kg)												
*** Grand Total ***		1	26,920 (kg)												

=== Last Item ===

DAECHANG CORP.



Jeonpseong Kim CEO

We guarantee particular outlined in the specification for inventory sales.
This material has been fully killed and made by basic oxygen process.
Test Certificate is issued according to ISO 10474/EN 10204 3.1.
We hereby certify that the material herein has been made in accordance with the order and above specification.

* Position - T : Top, M : Middle, B : Bottom
* Tensile Test. Direction : 1/4T Transversal, Gauge Length : 50 mm(Round),
YP Method : Upper Point
* Impact Test. Direction : 1/4T Longitudinal, Full-Size : 10 mmX10 mm
* Division - L : Ladle Analysis
* Supply Condition : As-Rolled unless otherwise Heat Treated.



본 검사증명서 위·변조·사·문서 위·변조 시 28만원 제231조 29로 불이익을 당할 수 있으며 본 증서는 KOI AS의 관리이 없습니다.

Surveyor To :

T.C.BAE

Order No./계약번호 : 01S6907533

PO No./주문번호 : 2407-SM355A-SK1(P1)

Supplier /수출자 : POSCO INTERNATIONAL CORPORATION

Commodity /품명 : PLATE

Customer /고객사 : SUNKYUNG STEEL

Spec & Type : KS D3515 SM355A

Size/치수		Product No. 제품번호	Quantity 수량	Weight 중량 (kg)	Heat No. 제강번호	Country of Melt & Pour	Position	Tensile Test 인장시험 YP TS EL (MPa) (%)			Impact Test 충격시험 V Notch +20 °C Energy (Joule)	Division	Chemical Composition/화합성분 C (%) Si (%) Mn (%) P (%) S (%) CEQ (%) PCM (%)						
85x2438x6096		PB98029102	1	9,915	SB07248	KOR	T	344	531	35	1: 224 2: 212 3: 248 A: 228	L	0.1533	0.383	1.434	0.0131	0.0044	0.41	0.25
*** Sub Total (Q10) ***			1	9,915 (kg)		* B/L Date ==> 2024/08/08							CEQ (G) = C + Mn/6 + (Ni + Cu)/15 + (Cr + Mo + V)/5 PCM (B) = C + Si/30 + (Mn + Cu + Cr)/20 + Ni/60 + Mo/15 + V/10 + S*5						
95x2438x6096		PB98029301	1	11,083	SB07248	KOR	T	344	531	35	1: 211 2: 257 3: 224 A: 231	L	0.1533	0.383	1.434	0.0131	0.0044	0.41	0.25
95x2438x6096		PB98029401	1	11,083	SB07248	KOR	T	344	531	35	1: 211 2: 257 3: 224 A: 231	L	0.1533	0.383	1.434	0.0131	0.0044	0.41	0.25
*** Sub Total (Q20) ***			2	22,166 (kg)		* B/L Date ==> 2024/08/08							CEQ (G) = C + Mn/6 + (Ni + Cu)/15 + (Cr + Mo + V)/5 PCM (B) = C + Si/30 + (Mn + Cu + Cr)/20 + Ni/60 + Mo/15 + V/10 + S*5						
*** Grade Total ***			3	32,081 (kg)															
*** Grand Total ***			3	32,081 (kg)															
												Last Item ==							
												DAECHANG COR							
																			

=== Last Item ===

DAECHANG CORP.

Jeon Seong Kim
CEO

This material has been fully killed and made by basic oxygen process. This material is fine-grained steel.
This material has been made by vacuum degassing process.
Test Certificate is issued according to ISO 10474/EN 10204 3.1.
We, POSCO, hereby certify that the products specified in the present MTC have been fully processed in South Korea. (Facility and address information is stated below)
We hereby certify that the material herein has been made in accordance with the order and above specification.

* Position - T : Top, M : Middle, B : Bottom
* Tensile Test Direction : 1/4T Transversal, Gauge Length : 50 mm(Round),
YP Method : Upper Point
* Impact Test Direction : 1/4T Longitudinal, Full-Size : 10 mmX10 mm
* Division - L : Ladle Analysis
* Supply Condition : As-Rolled unless otherwise Heat Treated.



본 김사영영서 위변조시 사문서 위조28형법 제231조2항로 불이익을 당할 수 있으며, 본 증서는 KOLAS와 관련이 없습니다.

Surveyor To :

poscomtc@posco.com
+82-2-3457-0114

S.H. Park

Order No./계약번호 : 01S6997199

Supplier : eSteel4U
/수문자

PO No./주문번호 : 01S6997199

Commodity : PLATE
/품명Customer : ESTEEL4U
/고객사Spec & Type : KS-SM355A
/규격

Size/치수	Product No. /제품번호	Quantity /수량	Weight /중량 (kg)	Heat No. /제강번호	Country of Melt & Pour	Position	Tensile Test /인장시험 YP TS EL (MPa) (%)	Impact Test /충격시험 V Notch +20 °C Energy (Joule)	Division	Chemical Composition/화합성분
170x2622x11000	PP54523801	1	38,490	SU231/2	KOR	T	355 560 32	1: 171 2: 156 3: 78 A: 35	L	C 0.1557 Si 0.387 Mn 1.421 P 0.0098 S 0.0031 CEQ 0.41 PCM 0.25
1 ... Sub Total (010) ...		1	38,490 (kg)							CEQ (G) = C + Mn/6 + (Ni + Cu)/15 + (Cr + Mo + V)/5 PCM (B) = C + Si/30 + (Mn + Cu + Cr)/20 + Ni/60 + Mo/15 + V/10 + 5*B
1 ... Grade Total ...		1	38,490 (kg)							
1 ... Grand Total ...		1	38,490 (kg)					=== Last Item ===		

DAECHANG CORP.



Jeongseong Kim
CEO

DAECHANG CORP.

Jeongseong Kim CEO

* Position - T : Top, M : Middle, B : Bottom
* Tensile Test. Direction : 1/4T Transversal, Gauge Length : 50 mm (Round),
YP Method : Upper Point
* Impact Test. Direction : 1/4T Longitudinal, Full-Size : 10 mmX10 mm
* Division - L : Ladle Analysis
* Supply Condition : As-Rolled unless otherwise Heat Treated.

We guarantee particular outlined in the specification for inventory sales.
This material has been fully killed and made by basic oxygen process.
Test Certificate is issued according to ISO 10474/EN 10204 3.1.
We, POSCO, hereby certify that the products specified in the present MTC have been fully processed in South Korea. (Facility and address information is stated below)
We hereby certify that the material herein has been made in accordance with the order and above specification.

본 검사증서 위 변조시 사문서 위조28형집 제231조29로 불이만을 당할 수 있으며, 본 증서는 KOLAS와 관련이 없습니다.

poscomtc@posco.com
+82-2-3457-0114

Surveyor To :

Kang, Deokgu

검사증명서
MILL TEST CERTIFICATE



HYUNDAI
STEEL

포항공장 : 경상북도 포항시 남구 동해안로 6363
6363, Donghaean-ro, Nam-gu, Pohang-si,
Gyeongsangbuk-do, Korea

주문번호
Order No.

종명
Commodity

제품규격
Specification

: D230918644

: Round Bar(봉강)

: KS D 3752 SM45C (JIS G 4051 S45C)

고객사
Customer

주문자
Contractor

: 대륜특수강(주)

: 대륜특수강(주)

증명서 번호
Certificate No.

발행일자
Date of Issue

: DO20230900543 [2 / 4]

: 2023-09-20

제강번호 Heat No.	제품번호 Product No.	제품치수 Dimension (mm)		수량 Quantity (PCS/BD)	중량 Weight (KG)	열처리 Heat Treatment	구분 Division	화합성분 Chemical Composition(%)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
		Diameter	Length					C	Si	Mn	P	S	Cu	N	Cr	N-O	단위 Unit				2 : x100				3 : x1,000				4 : x10,000				5 : x100,000																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													

구분 Division	압연비 Reduction Ratio	초음파 탐상검사 Ultrasonic Inspection	금속 Microscopic Inspection
Mh.			
Max.			
Result	20S		

N : MELTING PROCESS : EAF+LF+VD
O : SURFACE CONDITION : BLACK SURFACE
T : NO REPAIR WELDING
E : FULLY KILLED STEEL

DAECHANG CORP.

Jeongseong Kim

CEO

Signature

정영성

Chief Of Quality Assurance Team

WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS.

SeAH Besteel

522, Ochang-ro, Gunan-si, Jeollabuk-do, Republic of Korea (54007)

INSPECTION CERTIFICATE

ACCORDING TO EN10204:2004, Type 3.1



Date : 2023-07-20
Cert. No. : 202307-006343
Made in Korea
Melted and poured in the Republic of Korea

Steel Grade : SM45C (S45C)
Heat No. : D19974
Size : 60 mm
Commodity : Round Bar

Specification : KSD 3752 (JIS G 4051)
Melting Process : EAF-LF-VD(RH)-CC
Delivery Condition : As Rolled

Customer : (주)진흥특수강
Supplier : SeAH Besteel
Serial-No. :

Chemical Composition (wt. %)															Bundle	Quantity (pcs)	Weight (kg)	Length (mm)					
Inspection Items		C	Si	Mn	P	S	Cu	Ni	Cr	Mo	Ni+Cr	Product Hardness		Ultrasonic Test					Surface Inspection				
		×100	×100	×100	×1000	×1000	×100	×100	×100	×100	×100												
		Min.	42	15	60																		
		Max.	48	35	90	30	35	30	20	20										35			
		Result	45	18	66	18	12	11	5	16	2									21			
Inspection Items		Tensile Test				Charpy Impact Test				Product Hardness				Ultrasonic Test	Surface Inspection								
		Type				Type				Type													
		Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.										
		Spec.	Result	Spec.	Result	Spec.	Result	Spec.	Result	Spec.	Result	Spec.	Result										
		Result																					
Inspection Items		Gas Analysis (ppm)				Austenite Grain Size		Macro Structure				Macro Streak Flaw				Ultrasonic Test	Surface Inspection						
		Type				Type		Type				Type											
		Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.										
		Spec.	Result	Spec.	Result	Spec.	Result	Spec.	Result	Spec.	Result	Spec.	Result										
		Result																					
Inspection Item		Nonmetallic Inclusion														Reduction Ratio							
		Type																					
		Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.								
		Spec.	Result	Spec.	Result	Spec.	Result	Spec.	Result	Spec.	Result	Spec.	Result	Spec.	Result								
		Result																					
Inspection Items		Hardenability																					
		Type																					
		Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.								
		Spec.	Result	Spec.	Result	Spec.	Result	Spec.	Result	Spec.	Result	Spec.	Result	Spec.	Result								
		Result																					

DAECHANG CO

DAECHANG CO

Jeongseong Kim

DAECHANG CORP.

Jeongseong Kim
CEO

We hereby certify that the material described herein has been made in accordance with the rules of the contract.
Our quality system is certified in accordance with IATF 16949:2016 / ISO 9001:2015 by KSA
본 품질보증서에 명시된 규격용도의 사용시 안전상 문제가 발생할 수 있으며, 품질보증서 위·변조시 사문서 위조(형법231조)로 불이익을 당하실 수 있습니다.
TEL : +82-(0)63-460-8573(Repres.) / 8285(QA) / 8470(Technology Service) FAX : +82-(0)63-460-8423

Certified by S. J. Yoon.
Manager of Quality Assurance Team

SeAH Besteel

522, Ochang-ro, Gyeongsang-si, Jeollabuk-do, Republic of Korea (54007)

INSPECTION CERTIFICATE

ACCORDING TO EN10204:2004, Type 3.1

Date : 2023-04-13
Cert. No. : 202304-004762

Made in Korea

Melted and poured in the Republic of Korea



Page (1/1)

Steel Grade : SM45C (S45C)

Heat No. : B75267

Size : 80 mm

Commodity : Round Bar

Specification : KSD 3752 (JIS G 4051)

Melting Process : EAF-IF-VD(RID)-CC

Delivery Condition : As Rolled

Customer : (주)연동철수강

Supplier : SeAH Besteel

Serial No. :

Chemical Composition (wt. %)															
Inspection Items	C	Si	Mn	P	S	Cu	Ni	Cr	Mo	Ni+Cr	Bundle	Quantity (pcs)	Weight (kg)	Length (mm)	
	×100	×100	×100	×1000	×1000	×100	×100	×100	×100	×100					
Spec.	Min.	42	15	60	35	30	20	20	35	35		7	1,668	6,000	
	Max.	48	35	90	30	30	20	20	35	35		7	1,668	6,000	
	Result	46	20	65	16	16	12	4	12	1	16				
Inspection Items	Tensile Test				Charpy Impact Test				Product Hardness		Ultrasonic Test	Surface Inspection			
	Type		Type		Type		Type								
Spec.	Min.											GOOD	GOOD		
	Max.											GOOD	GOOD		
	Result														
Inspection Items	Gas Analysis(ppm)		Austenite Grain Size		Macro Structure		Macro Streak Flaw		Decarburized Layer						
	Result	Result	Result	Result	Result	Result	Result	Result							
Spec.	Min.														
	Max.														
	Result														
Inspection Item	Nonmetallic Inclusion												Reduction Ratio		
	Result														
Spec.	Min.														
	Max.														
	Result														
Inspection Items	Hardenability														
	Result														
Spec.	Min.														
	Max.														
	Result														

DAECHANG C

330

Jeongseon Kim

S&AH Besteel

522, Ocheung-ro, Gwangju-si, Jeollabuk-do, Republic of Korea (540077).

Steel Grade : SM45C (S45C)
Heat No. : D15004
Size : 100 mm
Commodity : Round Bar

INSPECTION CERTIFICATE

ACCORDING TO EN10204:2004, Type 3.1

Specification : KSD 3752 (JIS G 4051)
Melting Process : EAF-LF-VD(RH)-CC
Delivery Condition : As Rolled

Date : 2023-05-31
Cert. No. : 202305-010110

Made in Korea

Melted and poured in the Republic of Korea

Customer : (주)진흥복수강

Supplier : SeAH Besteel

Serial-No. :

Inspection Items		Chemical Composition (wt. %)										Tensile Test		Charpy Impact Test		Product Hardness		Ultrasonic		Surface Inspection		Bundle		Quantity (pcs)		Weight (kg)		Length (mm)			
		C	Si	Mn	P	S	Cu	Ni	Cr	Mo	Ni+Cr																				
Spec.	Min.	42	15	60	×1000	×1000	×100	×100	×100	×100	×100	Type		Macro Structure		Macro Streak Flaw		GOOD		GOOD		G002		6		2,226		6,000			
	Max.	48	35	90	30	35	30	20	20	1	35											G003		6		2,226		6,000			
	Result	45	21	86	18	11	13	6	17	1	23											G004		6		2,228		6,000			
Spec.	Min.													Macro Structure		Macro Streak Flaw		GOOD		GOOD		TOTAL		18		6,680		6,000			
	Max.																														
	Result																														
Inspection Item		Nonmetallic Inclusion										Hardenability		Hardenability		Hardenability		Hardenability		Hardenability		Hardenability		Hardenability		Hardenability		Hardenability			
Spec.	Min.																														
	Max.																														
	Result																														
Spec.	Min.																														
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	Result																														

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DAECHANG CORP.

JeonGyeong Mill CEO

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Our quality system is certified in accordance with IATF 16949:2016 / ISO 9001:2015 by KSA

본 품질보증서에 명시된 규격용도의 사용시 안전상 문제가 발생할 수 있으며, 품질보증서 위변조시 사문서 위조(형법211조)로 불이익을 당하실 수 있습니다.
TEL : +82-(0)63-460-8573(Repres.) / 8285(QA) / 8470(Technology Service) FAX : +82-(0)63-460-8423

Certified by S. J. Yoon.

Manager of Quality Assurance Team

INSPECTION CERTIFICATE

ACCORDING TO EN10204:2004, Type 3.1

Date : 2024-12-10

Cert. No. : 202412-002421

Made in Korea

Melted and poured in the Republic of Korea



Steel Grade : SM45C (S45C)

Heat No. : D25354

Size : 40 mm

Commodity : Round Bar

Specification : KS D 3752 (JIS G 4051)

Melting Process : EAF-F-V(D(RH))-CC

Delivery Condition : As Rolled

Customer : (주)유진중합

Supplier : SeAH Besteel

Serial-No. :

Inspection Items		Chemical Composition (wt. %)										Tensile Test		Charpy Impact Test		Product Hardness		Ultrasonic Test		Surface Inspection		Bundle		Quantity (pcs)		Weight (kg)		Length (mm)	
		C	Si	Mn	P	S	Cu	Ni	Cr	Mo	Ni+Cr																		
		×100	×100	×100	×1000	×1000	×100	×100	×100	×100	×100																		
Spec.	Min.	42	15	60	30	35	30	20	20	20	35																		
	Max.	48	35	90	30	35	30	20	20	20	35																		
Result		45	19	65	18	12	16	8	17	2	24																		
Inspection Items		Gas Analysis(ppm)										Austenite Grain Size		Macro Structure		Macro Streak Flaw		Decarburized Layer		DM-T									
		Type																											
Spec.	Min.																												
	Max.																												
Result																													
Inspection Item		Nonmetallic Inclusion																										Reduction Ratio	
Spec.	Min.																												
	Max.																												
Result																													
Inspection Items		Hardenability																											
Spec.	Min.																												
	Max.																												
Result																													

DAECHAN

<<Remark>>

DAECHANG CORP.

CEO

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Our quality system is certified in accordance with IATF 16949:2016 / ISO 9001:2015 by KSA

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TEL : +82-(0)63-460-8573 (Reprints.) / 8285(OA) / 8470(Technology Service) FAX : +82-(0)63-460-8423

Certified by

S. J. Yoon.

Manager of Quality Assurance Team

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발행일자
Date of Issue

恒常

TABLE CONTINUED

Chief of Quality Assurance Team

S&AH Besteel

522, Oehang-ro, Gusan-si, Jeollabuk-do, Republic of Korea (54007)

INSPECTION CERTIFICATE

ACCORDING TO EN10204:2004, Type 3.1

Steel Grade : SM45C (S45C)
Heat No. : 804652
Size : 70 mm
Commodity : Round Bar

Specification : KS D 3752 (JIS G 4051)
Melting Process : EAF-LF-VD(RH)-CC
Delivery Condition : As Rolled

Customer : (주)진흥물수강
Supplier : SeAH Besteel
Serial-No. :

Page (1/1)



Date : 2023-07-20

Cert. No. : 202307-006420

Made in Korea

Melted and poured in the Republic of Korea

Chemical Composition (wt. %)

Inspection Items	Chemical Composition (wt. %)										Bundle	Quantity (pcs)	Weight (kg)	Length (mm)
	C	Si	Mn	P	S	Cu	Ni	Cr	Mo	NH-Cr				
	×100	×100	×100	×1000	×1000	×100	×100	×100	×100	×100				
Min.	42	15	60											
Max.	48	35	90	30	35	30	20	20	2	35				
Result	47	22	70	12	20	7	8	10	2	18				

Charpy Impact Test

Inspection Items	Tensile Test		Type		Product Hardness		Ultrasonic Test		Surface Inspection		Bundle	Quantity (pcs)	Weight (kg)	Length (mm)
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.				
Result														

Gas Analysis (ppm)

Inspection Items	Austenite Grain Size		Macro Structure		Macro Strain Flow		Decarburized Layer	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Result								

Nonmetallic Inclusion

Inspection Item	Reduction Ratio	
	Min.	Max.
Result		

Hardenability

Inspection Items	Hardenability	
	Min.	Max.
Result		

<<Remark>>

DAECHANG CORP.

SeAH Besteel
Jinhyeong Kim
CEO

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Certified by S.J. Yoon.

Manager of Quality Assurance Team

검사증명서
MILL TEST CERTIFICATE

HYUNDAI
STEEL

포항공장 : 경상북도 포항시 남구 동해안로 6363
6363, Donghaean-ro, Nam-gu, Pohang-si,
Gyeongsangbuk-do, Korea



주문번호 Order No.	: D230918644	고객사 Customer	: 대원특수강(주)	증명서 번호 Certificate No.	: DO20230900543 [1 / 4]
종류 Commodity	: Round Bar(봉강)	주문자 Contractor	: 대원특수강(주)	발행일자 Date of Issue	: 2023-09-22
제품규격 Specification	: KS D 3752 SM45C (JIS G 4051 S45C)				

제강번호 Heat No.	제품번호 Product No.	제품치수 Dimension (mm)		수량 Quantity (PCS/BD)	중량 Weight (KG)	열처리 Heat Treatment	구분 Division	화학성분 Chemical Composition(%)																단위 Unit																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
		Diameter	Length					C	Si	Mn	P	S	Qu	N	O	N-O							2:x100	3:x1,000	4:x10,000	5:x100,000																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
4G7902	BM6203304A	55	6,200	22	2,544	HOT ROLLED	Mn	42	15	60																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															

WE HEREBY CERTIFY THAT THE MATERIAL HAS BEEN MADE AND TESTED IN ACCORDANCE WITH THE ABOVE SPECIFICATION AND THE REQUIREMENTS.	Signature 정영민
	Official Quality Assurance Team

DAECHANG CORP.

Jeongseong Kim CEO

* 해당제품은 검사증명서를 사본으로 제출하지 않습니다. 사본의 경우, 원제품과 동일하지 않을 수 있습니다. 문의: 02-772-7125

INSPECTION (TEST) REPORT

조질 검사 (시험) 성적서



0250220044

검사일자 : 2025-02-24

발행일자 : 2025-02-24

CUSTOMER 발 주 처	(주)영산볼트	품명	육각	품번	YS32BTBB	WEIGHT 중량	300 KG
		규격	10.9T(기본)	재질	4140	LOT NO	250222114P

SKETCH
요약도면

BLANK

업체LOT : 33*90.85.80.75.95

ITEM OF INSP 검 사 항 목	SPEC REQUIRED 요 구 사 항	RESULT 측 정 결 과					
		1	2	3	4	5	평 균
SURFACE HARDNESS 표면경도	HV 336~372						
CORE HARDNESS 내부경도	HRC 34~38	34.1	34.6	35	34.3	34.9	34.6
CUSTOMER CHECK 수요자 확인	HV 336~372						
	HRC 34~38						
STRUCTURE 조직	SOLBITE	양 호					
SURFACE CONDITION 외관	균열, 해로운 흠없을것	양 호					
DECABURIZATION 탈탄	표면부탈탄없을것	양 호					
특 기 사 항							

판 정

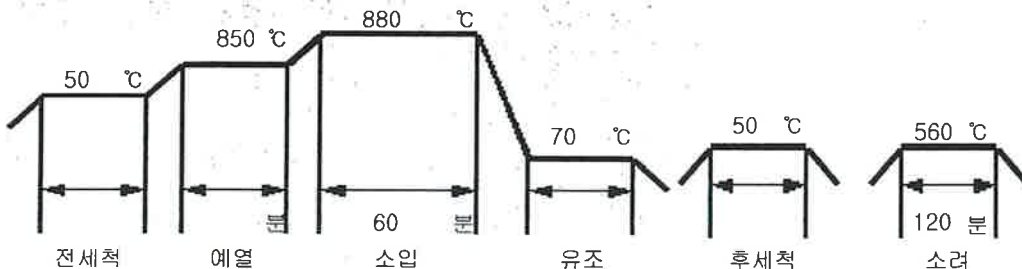
합격

2025-02-24

대광기업사
품질관리

대광기업사 품질관리

서동현



SeAH Besteel

522, Ocheung-ro, Guro-gu, Seoul, Republic of Korea (S-4007)

INSPECTION CERTIFICATE

ACCORDING TO EN10204:2004, Type 3.1

Steel Grade : S-SCM440
Heat No. : D26886
Size : 55 mm
Commodity : Round Bar

Specification :
Melting Process : EAF-LF-VDRH-CC
Delivery Condition : As Rolled

Date : 2025-04-02
Cert. No. : 202504-000007
Made in Korea
Melted and poured in the Republic of Korea
Customer : 주식회사 에이엘텍수강
Supplier : SeAH Besteel
Serial-No. :

Chemical Composition (wt. %)

Inspection Items	C	Si	Mn	P	S	Cu	Ni	Cr	Mo
Spec. Min.	×100	×100	×100	×1000	×1000	×100	×100	×100	×100
Spec. Max.	38	15	60	30	30	30	25	120	15
Result	43	35	90	28	13	18	9	93	16

Tensile Test

Inspection Items	Yield Strength	Tensile Strength	Elongation
Spec. Min.			
Spec. Max.			
Result			

Charpy Impact Test

Inspection Items	Energy
Spec. Min.	
Spec. Max.	
Result	

Product Hardness

Inspection Items	Hardness
Spec. Min.	
Spec. Max.	
Result	

Gas Analysis(ppm)

Inspection Items	Gas Analysis
Spec. Min.	
Spec. Max.	
Result	

Austenitic Grain Size

Inspection Items	Grain Size
Spec. Min.	
Spec. Max.	
Result	

Macro Structure

Inspection Items	Structure
Spec. Min.	
Spec. Max.	
Result	

Macro Streak Flaw

Inspection Items	Flaw
Spec. Min.	
Spec. Max.	
Result	

Decarburized Layer

Inspection Items	Layer
Spec. Min.	
Spec. Max.	
Result	

Nonmetallic Inclusion

Inspection Items	Inclusion
Spec. Min.	
Spec. Max.	
Result	

Hardenability

Inspection Items	Hardenability
Spec. Min.	
Spec. Max.	
Result	

<<Remark>>

We hereby certify that the material described herein has been made in accordance with the rules of the contract.
Our quality system is certified in accordance with IATF 16949:2016 / ISO 9001:2015 by KSA
* 단 품질보증서에 열거된 규격명도의 시험시 안전성 문제가 발생할 수 있으며, 품질보증서 유언조서 등에서 위조(偽造)될 수 있습니다.
TEL : +82-(0)2-6970-2192(Sales) +82-(0)63-460-8573(QA) / 8785(QA) / 8516(Technology Service) FAX : +82-(0)63-460-8423

Certified by

S. J. Yoon.

Manager of Quality Assurance Team.

DAECHANG CORP.

Jeongseong Kim CEO

원본 대조필

담당자: 나호준
연락처: 010-3378-4584

비밀번호 146
(경상남도 창원시 마산합포구 마산대로 336 102-1호)
주식회사 에이원텍수강

SCM440 55*211
SCM440 55*231

10/15



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검사시험성적서(Inspection Test Sheet)

한백열처리(HANBAK HEAT TREATMENT)

TEL : (031)319 - 1986 FAX : 319 - 1985

발행일 : 2025년 11월 04일

업 체 명 Order	(株) 永 山 BOLT	규 격 Size	39 * 100					
LOT. NO.	A5-251101-01	제품사진 (스캐치)						
공 정 Line	QUENCHING & TEMPERING	52*120 50EA 56*120 30EA 56*140 35EA						
품 명 Item	육 각							
재 질 Material	SCM440							
수 량 . 중 량 Number Weight	90 EA							
재 질 이 력 History of material	-							
요 구 경 도 Hardness	HRC 34 ~ 38							
허 용 변 형 Approved Distortion	변형 없을것							
경 화 층 Hardening Depth	-							
검 사 항 목 Inspection Item	기 준 치 Criterion	검 사 결 과 Actual Value					판 정 Decision	
		X ₁	X ₂	X ₃	X ₄	X ₅	평균	
외 관 Appearance	해로운 흠이 없을 것	- 양호 -						양 호
변 형 Distortion	mm	- 양호 -						양 호
표 면 경 도 Hardness	HRC 34 ~ 38	35	35.5	36	35	35.5		양 호
경 화 층 Hardening Depth	-	-						양 호
내 부 경 도 Hardness	-	-						-
경화층 측정 DATA								
mm								
HV								
						참고사항 Remarks		
						종합판정 		
한 백 열 처 리				(株) 永 山 BOLT				
검 사	검 토	승 인	담 당	검 토	승 인			
이 연 옥	정 연 태	박 연 규						

한 백 열 처 리 품 질 관 리 부

Jeongseong Kim

CEO

INSPECTION CERTIFICATE EN 10204-3.1

Customer: 신진화스너공업(주)/본점/
Order-No: 1002025100010092
Lot No: 1251027010178
Certificate No: D260116-92
Issue Date: 20260116
Description: □M20x2.5Px60 ◎SJF 10.9 (호칭 46S) 육각볼트/Alloy Steels 10.9/열처리방형
Delivery quantity: 4,430

Geometric properties (ISO 2859 AQL=1.0 S-1)

NO.	Characteristics	Standard (Min - Max)	Test results (Min - Max)	Samples	REJ
1	맞변거리(Width across flats)	29.6 ~ 30.0	29.71~29.81	3	0
2	머리부높이(Height of head)	12.65 ~ 13.0	12.88~12.9	3	0
3	호칭길이(Length)	59.3 ~ 60.0	59.85~59.92	3	0
4	나사부길이(Thread length)	48.5 ~ 51.0	49.87~49.96	3	0
5	원통부지름 (Body dia. Ds)	19.67 ~ 20.0	19.78~19.83	3	0
6	바깥지름(Major dia)	19.71 ~ 19.95	19.76~19.9	3	0
7	마크(Marking)	KS SJF 10.9	합격	3	0

Chemical composition (원재료 성분분석)

Material 51B20(국산)

Heat No.	C[%]	Mn[%]	P[%]	S[%]	Si[%]	Cr[%]	Mo[%]	V[%]	Ni[%]	B[%]	Cu[%]	Ti[%]	Al[%]
SF95651	0.21	0.74	0.008	0.005	0.19	0.73	0.	0.	0.	0.0019	0.	0.	0.

Mechanical properties (기계적 성질)

NO.	Characteristics	Standard (Min - Max)	Test results (Min - Max)	Samples	REJ
1	경도(Hardness)	HRC 33 ~ 39	36.4	1	0
2	인장강도(Tensile strength)	Min 1040 N/mm ²	1125	1	0
3	***항복강도(Yiele strength)	Min 940 N/mm ²	1012	1	0

Inspection result: Samples tested conformed to all of the specifications as above.



Prepared by ohj

Q.A. SIGNATORY

- 이 시험성적서는 KS Q ISO/IEC 17025 및 KOLAS 인정과 관련이 없음을 밝힙니다.
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DAECHANG CORP.

Jeongseong Kim CEO

INSPECTION CERTIFICATE EN 10204-3.1

Customer: 정기발주(통합)
Order-No: 1002025111500010026
Lot No: 1251201010222
Certificate No: D260116-94
Issue Date: 20260116
Description: M20x2.5Px70 SJF 10.9 (호칭 46S) 육각볼트/Alloy Steels 10.9/열처리방청
Delivery quantity: 8,005

Geometric properties (ISO 2859 AQL=1.0 S-1)

NO.	Characteristics	Standard (Min - Max)	Test results (Min - Max)	Samples	REJ
1	맞변거리(Width across flats)	29.6 ~ 30.0	29.72~29.83	3	0
2	머리부높이(Height of head)	12.65 ~ 13.0	12.89~12.9	3	0
3	호칭길이(Length)	69.3 ~ 70.0	69.45~69.79	3	0
4	나사부길이(Thread length)	48.5 ~ 51.0	48.57~50.38	3	0
5	원통부지름 (Body dia. Ds)	19.65 ~ 20.0	19.77~19.92	3	0
6	바깥지름(Major dia)	19.71 ~ 19.95	19.77~19.79	3	0
7	마크(Marking)	KS SJF 10.9	합격	3	0

Chemical composition (원재료 성분분석)

Material 51B20(국산)

Heat No.	C[%]	Mn[%]	P[%]	S[%]	Si[%]	Cr[%]	Mo[%]	V[%]	Ni[%]	B[%]	Cu[%]	Ti[%]	Al[%]
SF95651	0.21	0.74	0.008	0.005	0.19	0.73	0.	0.	0.	0.0019	0.	0.	0.

Mechanical properties (기계적 성질)

NO.	Characteristics	Standard (Min - Max)	Test results (Min - Max)	Samples	REJ
1	경도(Hardness)	HRC 33 ~ 39	36.8	1	0
2	인장강도(Tensile strength)	Min 1040 N/mm ²	1150	1	0
3	"*"항복강도(Yiele strength)	Min 940 N/mm ²	1035	1	0

Inspection result: Samples tested conformed to all of the specifications as above.



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Prepared by ohj

DAECHANG CORP.

Jeongseong Kim

CEO

INSPECTION CERTIFICATE EN 10204-3.1

Customer: (주)신진화스너/부산대저
Order-No: 1002025112500200003
Lot No: 1251215010048
Certificate No: D260116-96
Issue Date: 20260116
Description: M20x2.5Px80 SJF 10.9 (호칭 46S) 육각볼트/Alloy Steels 10.9/열처리방청
Delivery quantity: 3,515

Geometric properties (ISO 2859 AQL=1.0 S-1)

NO.	Characteristics	Standard (Min - Max)	Test results (Min - Max)	Samples	REJ
1	맞변거리(Width across flats)	29.6 ~ 30.0	29.71~29.84	3	0
2	머리높이(Height of head)	12.65 ~ 13.0	12.81~12.9	3	0
3	호칭길이(Length)	79.3 ~ 80.0	79.8~79.86	3	0
4	나사부길이(Thread length)	48.5 ~ 51.0	49.7~49.81	3	0
5	원통부지름 (Body dia. Ds)	19.65 ~ 20.0	19.76~19.84	3	0
6	바깥지름(Major dia)	19.71 ~ 19.95	19.78~19.9	3	0
7	마크(Marking)	KS SJF 10.9	합격	3	0

Chemical composition (원재료 성분분석)

Material 51B20(국산)

Heat No.	C[%]	Mn[%]	P[%]	S[%]	Si[%]	Cr[%]	Mo[%]	V[%]	Ni[%]	B[%]	Cu[%]	Ti[%]	Al[%]
SF98284	0.213	0.733	0.0126	0.006	0.18	0.722	0.	0.	0.	0.0017	0.	0.	0.

Mechanical properties (기계적 성질)

NO.	Characteristics	Standard (Min - Max)	Test results (Min - Max)	Samples	REJ
1	경도(Hardness)	HRC 33 ~ 39	36.9	1	0
2	인장강도(Tensile strength)	Min 1040 N/mm ²	1155	1	0
3	"**"항복강도(Yiele strength)	Min 940 N/mm ²	1039	1	0

Inspection result: Samples tested conformed to all of the specifications as above.



Prepared by ohj

Q.A. SIGNATORY

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DAECHAN CORP.

Jeongseong Kim

CEO

INSPECTION CERTIFICATE EN 10204-3.1

Customer: (주)신진화스너/부산대저
Order-No: 1002025100010164
Lot No: 1251110010186
Certificate No: D260116-53
Issue Date: 20260116
Description: □M24x3.0Px70 ◎SJF 10.9 (호칭 54S) 육각볼트/Alloy Steels 10.9/열처리방청
Delivery quantity: 2,405

Geometric properties (ISO 2859 AQL=1.0 S-1)

NO.	Characteristics	Standard (Min - Max)	Test results (Min - Max)	Samples	REJ
1	맞변거리(Width across flats)	35.6 ~ 36.0	35.61~35.66	3	0
2	머리부높이(Height of head)	14.65 ~ 15.0	14.83~14.94	3	0
3	호칭길이(Length)	69.5 ~ 70	69.5~69.63	3	0
4	나사부길이(Thread length)	57 ~ 60.0	57.58~59.95	3	0
5	원통부지름 (Body dia. Ds)	23.65 ~ 24.0	23.77~23.78	3	0
6	바깥지름(Major dia)	23.68 ~ 23.94	23.73~23.75	3	0
7	마크(Marking)	KS SJF 10.9	합격	3	0

Chemical composition (원재료 성분분석)

Material 51B20(국산)

Heat No.	C[%]	Mn[%]	P[%]	S[%]	Si[%]	Cr[%]	Mo[%]	V[%]	Ni[%]	B[%]	Cu[%]	Ti[%]	Al[%]
SF94576	0.21	0.73	0.012	0.006	0.19	0.73	0.	0.	0.	0.002	0.	0.	0.

Mechanical properties (기계적 성질)

NO.	Characteristics	Standard (Min - Max)	Test results (Min - Max)	Samples	REJ
1	경도(Hardness)	HRC 33 ~ 39	36.6	1	0
2	인장강도(Tensile strength)	Min 1040 N/mm ²	1130	1	0
3	***항복강도(Yiele strength)	Min 940 N/mm ²	1017	1	0

Inspection result: Samples tested conformed to all of the specifications as above.



Prepared by hjh

Q.A. SIGNATORY

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DAECHANG CORP.

Jongsong Kim
CEO

INSPECTION CERTIFICATE EN 10204-3.1

Customer: (주)광명볼트/화성팔탄
Order-No: 1002025062000020005
Lot No: 1250721010088
Certificate No: D260422-02
Issue Date: 20260422
Description: M27x3.0Px70 SJF 10.9 전산 육각볼트/Alloy Steels 10.9/열처리방정
Delivery quantity: 1,190

Geometric properties (ISO 2859 AQL=1.0 S-1)

NO.	Characteristics	Standard (Min - Max)	Test results (Min - Max)	Samples	REJ
1	맞변거리(Width across flats)	40.0 ~ 41.0	40.33~40.69	3	0
2	머리부높이(Height of head)	16.65 ~ 17.0	16.72~16.85	3	0
3	호칭길이(Length)	69.4 ~ 70.0	69.82~69.99	3	0
4	바깥지름(Major dia)	26.68 ~ 26.94	26.79~26.88	3	0
5	마크(Marking)	KR KS SJF 10.9	합격	3	0

Chemical composition (원재료 성분분석)

Material	SCr 420B(D)												
Heat No.	C[%]	Mn[%]	P[%]	S[%]	Si[%]	Cr[%]	Mo[%]	V[%]	Ni[%]	B[%]	Cu[%]	Ti[%]	Al[%]
SF89529	0.225	0.765	0.0075	0.0061	0.189	0.964	0.	0.	0.016	0.0017	0.016	0.	0.

Mechanical properties (기계적 성질)

NO.	Characteristics	Standard (Min - Max)	Test results (Min - Max)	Samples	REJ
1	경도(Hardness)	HRC 33 ~ 39	36.8	1	0
2	인장강도(Tensile strength)	Min 1040 N/mm ²	1125	1	0
3	***항복강도(Yiele strength)	Min 940 N/mm ²	1039	1	0

Inspection result: Samples tested conformed to all of the specifications as above.



Prepared by hjh

Q.A. SIGNATORY

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DAECHANG CORP.

Jeongseong Kim

CEO

검 사 증 명 서

(TEST CERTIFICATE EN10204 TYPE 3.1)

한 양 화 스 니 공 업(주)

HanYang Fasteners Industry Inc.
경기도 화성시 정남면 서봉로 851번길 68-7
68-7, Seobong-ro 851beon-gil, Jeongnam-myeon
Hwaseong-si, Gyeonggi-do, KOREA
TEL) +82 31 366 2348 FAX) +82 31 366 2341

Customer : 현대볼트상사
Product Name : 렌지볼트(12.9 SCM435)
Specification : KS B1003 / KS B0233

Cert No. : HYF250612-01
Date OF ISSUE : 2025. 06. 12
P A G E : 1/1

품 명 PARTS NAME	규 격 NOMINAL SIZE	수 량 QUANTITY EA	로트번호 LOT NO HEAT NO	기 계 적 성 질 (MECHANICAL PROPERTIES)						토오크계수치		화학적성분% (CHEMICAL COMPOSITION)										비 고
				제품인장시험 FULL SIZE TEST			시 편 인 장 시 험 MACHINED SPECIMEN TEST			HARD NESS		TORQUE COEFIENCE K	C	Si	Mn	P	S	Ni	Cr	Cu	Mo	
				TL	PL	TS	YS	EX	RA	HRC	NESS											
BOLT	M16X30	220	G332C08321	205827	-	1311	1253	-	55	40	-	36	20	73	16	1	2	105	2	16		
BOLT	M16X40	150	G332C08321	206455	-	1315	1242	-	55	42	-	36	20	73	16	1	2	105	2	16		
BOLT	M20X45	100	G332B11524	323155	-	1319	1252	-	53	41	-	36	21	74	10	3	2	108	2	16		

NOTE. TL: TENSILE LOAD(인장하중:N), PL:PROOF LOAD(니트 보증하중:N), TS:TENSILE STRENGTH(인장강도:N/mm²), YS:YIELD STRENGTH(항복강도:N/mm²), EL: ELONGATION(연신율:%),
HRC:ROCKWELL HARDNESS TEST, RA:REDUCTION OF AREA(단면수축율:%), EX:EXTENSION(판단후 신장:mm), T(토오크)=K(토오크계수치)xD(볼트의 호칭경)N(축력), 9.8N=1Kgf

위의 제품들은 규정대로 생산, 시험을 행하였으며 사양에서 요구하는 모든 조건에 합격되었음을 증명합니다.

WE CERTIFY THE COMMODITY HAS BEEN MANUFACTURED AND TESTED IN ACCORDING WITH THE ABOVE MENTIONED SPECIFICATION, AND THAT ALL REQUIREMENTS OF THE SPECIFICATION.

Signature

Principal Manager of Quality Assurance Team (T.Y JOO)

HYF-QP-F103-3

한양화스너공업(주)

A4(210X297)

DAECHANG CORP.



Yongseong Kim CEO

검사증명서 (Mill Test Certificate)

수요자(Customer) : (주)드림에스티
 증명서 번호(Certification No.) : DIQ-2508-0929
 발행일자(Date of Issue) : 2025-08-22

Head Office : 111, Daesong-ro, Nam-gu,
 Pohang-si, Gyeongsangbuk-do
 TEL / FAX : 054)285-7251~4/054)285-1250
 Office / SEOUL 02)312-2201~4
 BUSAN : 051)441-6561~2
 TAEGU : 053)756-1201~5

제품번호 Product No	제품군 Product Category	제품규격 Standard	치수 Size	길이 Length	중량 Weight	본수 Quantity	제품번호 Product No	제강번호 Heat No
단위(Unit)			mm	mm	kg	PCS		
S60895-01	LA-ROUND BAR	AISI4140	30.00	6,500	4.494	124	S60895-01	S60895
합계(Total)					4.494	124		

제품번호 Product No	Chemical Composition														
Spec.	Max.	C	Si	Mn	P	S	Ni	Cr	Mo	Cu	Al				
	Min.	43	35	100	35	40		110	25						
단위(Unit)		X100	X100	X100	X1000	X1000	X100	X100	X100	X100	X1000				
S60895-01		41.6	24.7	82	13.2	12.7	9.8	105.7	19.3	10.8	13.9				

- End Line -

DAECHANG CORP.

Jeongseong Kim

CEO

Remark : WE HEREBY CERTIFY THAT THE MATERIAL HEREIN HAS BEEN MADE IN ACCORDANCE WITH THE ORDER AND SPECIFICATION. NO REPAIR WELDING. ORIGIN : MADE IN KOREA

Hyun Jik Lee

QUALITY ASSURANCE TEAM MANAGER



DONGIL INDUSTRIES CO., LTD.

동일산업(주) 봉강사업부

SeAH Besteel

522, Ochang-ro, Gunmn-si, Jollabuk-do, Republic of Korea (54007)

Chứng chỉ kiểm tra

ACCORDING TO EN10204:2004, Type 3.1

Date : 2026-01-02
Cert. No. : 202601-000379Made in Korea
Melted and poured in the Republic of Korea

Page (1/1)

Thép thô : SCM440
Số mẻ luyện : B90990
Kích cỡ : 45 mm
Hàng hóa : Thép trònThông số kỹ thuật : KS D 3867 (JIS G 4053)
Quá trình nấu chảy : EAF-LF-VD(RH)-CC
Tình trạng giao hàng : As RolledKhách hàng : 주식회사 에이원특수강
cung cấp : SeAH Besteel
Serial-No. :

Inspection Items		Chemical Composition (wt. %)															
		C	Si	Mn	P	S	Cu	Ni	Cr	Mo							
		×100	×100	×100	×1000	×1000	×100	×100	×100	×100							
Spec.	Min.	38	15	60					90	15							
	Max.	43	35	90	30	30	30	25	120	30							
Result		39	20	70	20	15	19	8	93	16							
I Mục kiểm tra		Thử nghiệm kéo				Thử nghiệm va đập Charpy				Độ cứng sản phẩm		Kiểm tra siêu âm	Surface Inspection	Bundle	Quantity (pcs)	Weight (kg)	Length (mm)
Spec.	Min.					Type								G008	26	1.954	
	Max.													TOTAL	26	1.954	
Result												GOOD	GOOD				
Inspection Items		Phân tích khí(ppm)			Kích thước hạt austenit		Cấu trúc thô đại			Khuyết tật vật thô đại			lớp khử carbon				
													DM-T				
Spec.	Min.																
	Max.																
Result																	
Mục kiểm tra		Không có tạp chất kim loại										Reduction Ratio					
Spec.	Min.																
	Max.																
Result																	
Mục kiểm tra		Khả năng tôi cứng															
Spec.	Min.																
	Max.																
Result																	

<<Remark>>

담당자: 나호준
연락처: 010-3378-4584

We hereby certify that the material described herein has been made in accordance with the rules of the contract.

Our quality system is certified in accordance with IATF 16949:2016 / ISO 9001:2015 by KSA

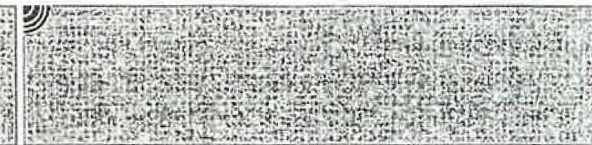
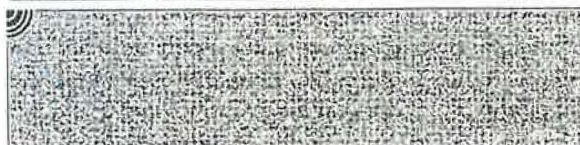
※ 본 품질보증서에 명시된 규격용도의 사용시 안전성 문제가 발생할 수 있으며, 품질보증서 위변조시 사문서 위조(형법23 조)로 불이익을 당할 수 있습니다.

TEL : +82-(0)2-6970-2192(Sales) +82-(0)63-460-8573(QA) / 8285(QA) / 8516(Technology Service) FAX : +82-(0)63-460-8423

Certified by

B.H. PARK

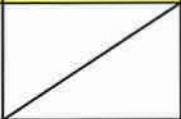
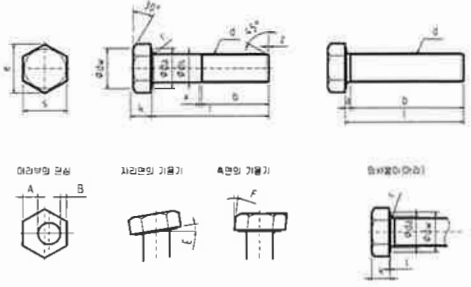
Manager of Quality Assurance Team.



Jongseong Kim

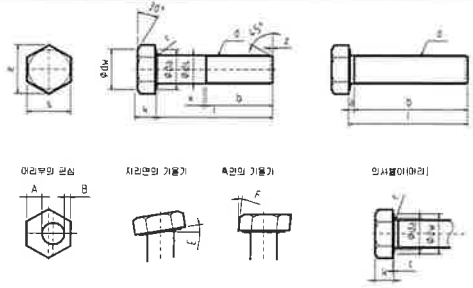
CEO

제품검사 성적서

종합 판정		제작업	수 정	특 채	발 행 처	작 성	검 토	승 인					
合格					 한양화스너공업(주)								
고객명	현대볼트상사												
품 명	51B20 강재 6각 볼트												
등 급	10.9	규격(d× l)	M10x1.25Px55										
로트번호	2026-03-24	로트크기	300 EA										
검사자	주태영	검사일자	2026년 3월 24일										
순서	검 사 항 목		검사방식	시료 크기	검사방법	합부판정기준	검 사 결 과						
							X1	X2	X3	X4	X5	합부	
1	기계적성질	인 장 강 도	체크검사 n = 1 c = 0	1	인장시험	최소 1040 N/mm2	1.135					합	
		0.2% 내력				최소 940 N/mm2	1.001					합	
		경도(HRC)				HRC 32 ~ 39	35.8					합	
2	모양·치수	호칭길이(l)	체크검사 n = 5 c = 0	5	버니어 캘리퍼스	55.0 ± 0.7mm	54.93	55.04	55.05	54.94	55.09	합	
		머리부높이(k)				7.0 ± 0.3 mm	7.07	7.02	7.08	7.00	7.01	합	
		맞변거리(s)				17 - 0.7 mm	16.93	16.90	16.95	16.92	16.94	합	
		대 각(e)				약 19.6 mm	19.54	19.57	19.53	19.58	19.55	합	
		머리부(dk')			투 영 기	약 16.5 mm	16.51	16.52	16.54	16.52	16.52	합	
		원통부지름(ds)			마이크로메타	10.0 - 0.20 mm	9.94	9.92	9.90	9.92	9.91	합	
		자리면지름(dw)			투 영 기	최소 15.8 mm	16.01	15.91	15.90	16.05	15.97	합	
		목밀부둥글기(r)				최소 0.40 mm	0.56	0.52	0.56	0.52	0.59	합	
		목밀부지름(da)				최대 11.2 mm	11.02	11.09	11.05	11.10	11.08	합	
		머리부편심(A-B)				최대 0.5 mm	0.18	0.19	0.16	0.18	0.18	합	
		자리면기울기(E)				최대 1°	0.16	0.18	0.17	0.19	0.19	합	
		측면기울기(F)				최대 2°	0.68	0.69	0.58	0.69	0.56	합	
3	나사	나사게이지			G0 게이지	한계게이지에적합	양호	양호	양호	양호	양호	합	
		나사게이지			N0 게이지	한계게이지에적합	양호	양호	양호	양호	양호	합	
4	겉모양	표면거칠기			자리면	조도측정기	25S	양호	양호	양호	양호	양호	합
							축 부	50S	양호	양호	양호	양호	양호
5	표시	표 면 결 함			육 안	해로운결함없을것	양호	양호	양호	양호	양호	합	
							머리부윗면에표시	양호	양호	양호	양호	양호	합
							포장바깥면에표시	양호	양호	양호	양호	양호	합



제품검사 성적서

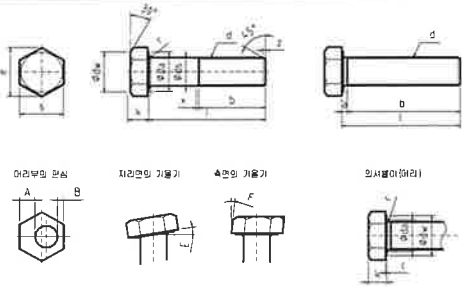
종합 판정		재작업	수 정	특 채	발 행 처	작 성	검 토	승 인				
合格					 한양하스너공업(주)							
고객명	현대볼트상사											
품 명	51B20 강재 6각 볼트 (전산)											
등 급	10.9	규격(d× l)	M12x1.75Px50									
로트번호	2026-03-24	로트크기	200 EA									
검사자	주태영	검사일자	2026년 3월 24일									
순서	검 사 항 목		검사방식	시료 크기	검사방법	합부판정기준	검 사 결 과					
							X1	X2	X3	X4	X5	합부
1	기계적성질	인 장 강 도	체크검사 n = 1 c = 0	1	인장시험	최소 1040 N/mm2	1,147					합
		0.2% 내력				최소 940 N/mm2	1,037				합	
		경도(HRC)				HRC 32~39	34.6				합	
2	모양·치수	호칭길이(l)	체크검사 n = 5 c = 0	5	버어니어 캘리퍼스	50.0 ± 0.25mm	49.99	49.96	49.98	50.01	50.01	합
		머리부높이(k)				8 ± 0.3 mm	7.93	7.95	7.91	7.90	7.95	합
		맞변거리(s)				19 - 0.8 mm	18.87	18.88	18.80	18.84	18.84	합
		대 각(e)				약 21.9 mm	21.86	21.85	21.88	21.89	21.88	합
		머리부(dk')			투영기	약 18 mm	17.59	17.54	17.52	17.56	17.50	합
		자리면지름(dw)				최소 17.6 mm	17.74	17.79	17.70	17.78	17.79	합
		목밀부둥글기(r)				최소 0.6 mm	0.70	0.70	0.67	0.68	0.67	합
		머리부편심(A-B)				최대 0.7 mm	0.27	0.26	0.28	0.20	0.25	합
		자리면기울기(E)				최대 1°	0.11	0.12	0.12	0.10	0.11	합
		측면기울기(F)				최대 2°	0.52	0.54	0.51	0.53	0.52	합
3	나사	나사게이지			G0 게이지	한계게이지에 적합	양호	양호	양호	양호	양호	합
		나사게이지			N0 게이지	한계게이지에 적합	양호	양호	양호	양호	양호	합
4	결모양	표면거칠기			조도측정기	25S	양호	양호	양호	양호	양호	합
						50S	양호	양호	양호	양호	양호	합
5	표시	표면결함			육안	해로운결함없을것	양호	양호	양호	양호	양호	합
						머리부윗면에표시	양호	양호	양호	양호	양호	합
		포장바깥면에표시			양호	양호	양호	양호	양호	합		

HYF-QP 양식 F103-2B Rev NO.0

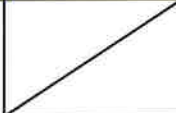
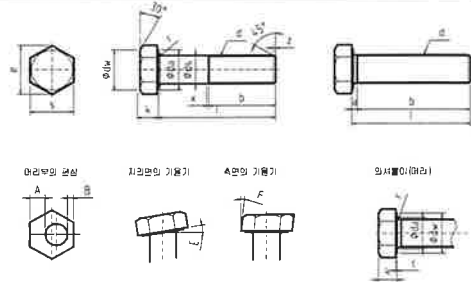
한양화스너공업(주) DAECHANG CORP. (A4210 X 297)

Jeongseong Kim CEO

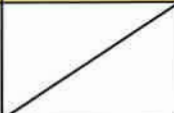
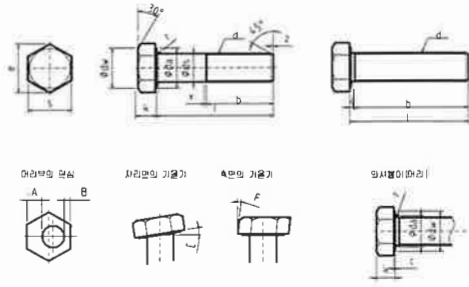
제품검사 성적서

종합 판정		재작업	수 정	특 채	발 행 처	작 성	검 토	승 인					
合格					 한양하스너공업(주)								
고객명	현대볼트상사												
품 명	51B20 강재 6각 볼트(전산)												
등 급	10.9	규격(d× l)	M12x1.75Px55										
로트번호	2026-03-24	로트크기	200 EA										
검사자	주태영	검사일자	2026년 3월 24일										
순서	검 사 항 목		검사방식	시료 크기	검사방법	합부판정기준	검 사 결 과						
							X1	X2	X3	X4	X5	합부	
1	기계적성질	인 장 강 도	체크검사 n = 1 c = 0	1	인장시험	최소 1040 N/mm2	1,114					합	
		0.2% 내력				최소 940 N/mm2	995					합	
		경도(HRC)				HRC 32~39	35.3					합	
2	모양·치수	호칭길이(l)	체크검사 n = 5 c = 0	5	버어니어 캘리퍼스	55.0 ± 0.35mm	55.03	55.07	55.02	54.93	55.05	합	
		머리부높이(k)				8 ± 0.3 mm	7.93	7.92	7.94	7.94	7.93	합	
		맞변거리(s)				19 - 0.8 mm	18.87	18.83	18.83	18.81	18.88	합	
		대 각(e)				약 21.9 mm	21.89	21.86	21.89	21.88	21.86	합	
		머리부(dk')			투 영 기	약 18 mm	17.55	17.60	17.60	17.50	17.59	합	
		원통부지름(ds)			마이크로메타	12 - 0.25 mm	11.85	11.88	11.91	11.92	11.91	합	
		자리면지름(dw)			투 영 기	최소 17.6 mm	17.78	17.75	17.74	17.77	17.79	합	
		목밀부동글기(r)				최소 0.6 mm	0.67	0.69	0.69	0.70	0.70	합	
		머리부편심(A-B)				최대 0.7 mm	0.21	0.23	0.22	0.24	0.23	합	
		자리면기울기(E)				최대 1°	0.11	0.11	0.10	0.12	0.10	합	
		측면기울기(F)			최대 2°	0.53	0.57	0.47	0.59	0.54	합		
3	나사	나사게이지			GO 게이지	한계게이지에적합	양호	양호	양호	양호	양호	합	
		나사게이지			NO 게이지	한계게이지에적합	양호	양호	양호	양호	양호	합	
4	결모양	표면거칠기			자리면	조도측정기	25S	양호	양호	양호	양호	양호	합
					측 부		50S	양호	양호	양호	양호	양호	합
		표 면 결 함			육 안	해로운결함없을것	양호	양호	양호	양호	양호	합	
표시	제 품	머리부윗면에표시				양호	양호	양호	양호	양호	합		
	포 장	포장바깥면에표시				양호	양호	양호	양호	양호	합		

제품검사 성적서

종합 판정		재작업	수 정	특 채	발 행 처	작 성	검 토	승 인				
合格					 한양하스너공업(주)							
고객명	현대볼트상사											
품 명	51B20 강재 6각 볼트 (전산)											
등 급	10.9	규격(d× l)	M20x1.5Px70									
로트번호	2026-03-24	로트크기	405 EA									
검사자	주태영	검사일자	2026년 3월 24일									
순서	검 사 항 목		검사방식	시료 크기	검사방법	합부판정기준	검 사 결 과					
							X1	X2	X3	X4	X5	합부
1	기계적성질	인 장 강 도	체크검사 n = 1 c = 0	1	인장시험	최소 1040 N/mm2	1.126					합
		0.2% 내력				최소 940 N/mm2	1.027					합
		경도(HRC)				HRC 32~39	36.1					합
2	모양·치수	호칭길이(l)	체크검사 n = 5 c = 0	5	버어니어 캘리퍼스	70 ± 0.7mm	70.02	69.95	69.99	70.05	70.07	합
		머리부높이(k)				13 ± 0.35 mm	13.05	13.09	13.09	13.09	13.09	합
		맞변거리(s)				30 - 0.8 mm	29.91	29.85	29.85	29.91	29.92	합
		대 각(e)				약 34.6 mm	34.55	34.55	34.53	34.55	34.54	합
		머리부(dk')			투영기	약 29 mm	28.93	28.90	28.93	28.94	28.95	합
		자리면지름(dw)			투영기	최소 28.5 mm	28.74	28.66	28.72	28.72	28.73	합
		목밀부둥글기(r)				최소 0.8 mm	0.90	0.93	0.93	0.93	0.90	합
		목밀부지름(da)				최대 22.4 mm	22.30	22.23	22.21	22.25	22.27	합
		머리부편심(A-B)				최대 0.9 mm	0.30	0.39	0.37	0.39	0.38	합
		자리면기울기(E)				최대 1°	0.15	0.10	0.12	0.11	0.11	합
		측면기울기(F)				최대 2°	0.57	0.69	0.66	0.62	0.63	합
3	나사	나사게이지			G0 게이지	한계게이지에적합	양호	양호	양호	양호	양호	합
		나사게이지			N0 게이지	한계게이지에적합	양호	양호	양호	양호	양호	합
4	겉모양	표면거칠기			조도측정기	25S	양호	양호	양호	양호	양호	합
		촉 부				50S	양호	양호	양호	양호	양호	합
		표면결함			육안	해로운결함없을것	양호	양호	양호	양호	양호	합
제 품	머리부윗면에표시	양호				양호	양호	양호	양호	합		
5	표시	포 장				포장바깥면에표시	양호	양호	양호	양호	양호	합

제품검사 성적서

종합 판정		재작업	수 정	특 채	발 행 처	작 성	검 토	승 인					
合格					 한양화스너공업(주)								
고객명	현대볼트상사												
품 명	51B20 강재 6각 볼트 (전산)												
등 급	10.9	규격(d×l)	M24x3.0Px80										
로트번호	2026-03-24	로트크기	30 EA										
검사자	주태영	검사일자	2026년 3월 24일										
순서	검 사 항 목		검사방식	시료 크기	검사방법	합부판정기준	검 사 결 과						
							X1	X2	X3	X4	X5	합부	
1	기계적성질	인 장 강 도	체크검사 n = 1 c = 0	1	인장시험	최소 1040 N/mm2	1.120						합
		0.2% 내력				최소 940 N/mm2	992						합
		경도(HRC)				HRC 32~39	34.5						합
2	모양 · 치수	호칭길이(l)	체크검사 n = 5 c = 0	5	베어니어 캘리퍼스	80 ± 0.7mm	80.07	80.07	80.06	79.93	79.96	합	
		머리부높이(k)				15 ± 0.35 mm	15.10	15.10	15.09	15.07	15.10	합	
		맞변거리(s)				36 - 1.3 mm	35.93	35.87	35.95	35.95	35.92	합	
		대 각(e)				약 41.6 mm	41.61	41.57	41.56	41.58	41.55	합	
		머리부(dk')			투영기	약 34 mm	34.04	34.04	34.04	33.97	33.98	합	
		자리면지름(dw)			투영기	최소 34.2 mm	34.50	34.52	34.56	34.50	34.57	합	
		목밑부둥글기(r)				최소 0.8 mm	0.91	0.92	0.90	0.95	0.92	합	
		목밑부지름(da)				최대 26.4 mm	26.26	26.25	26.21	26.20	26.29	합	
		머리부편심(A-B)				최대 1.2 mm	0.52	0.59	0.60	0.59	0.57	합	
		자리면기울기(E)				최대 1°	0.23	0.21	0.22	0.23	0.25	합	
		측면기울기(F)				최대 2°	0.61	0.68	0.68	0.66	0.69	합	
3	나사	나사게이지			GO 게이지	한계게이지에적합	양호	양호	양호	양호	양호	합	
		나사게이지			NO 게이지	한계게이지에적합	양호	양호	양호	양호	양호	합	
4	결 모양	표면거칠기			조도측정기	25S	양호	양호	양호	양호	양호	합	
		촉 부				50S	양호	양호	양호	양호	양호	합	
		표면 결 함			육안	헤로운결함없을것	양호	양호	양호	양호	양호	합	
제 품	머리부윗면에표시	양호				양호	양호	양호	양호	합			
포 장	포장바깥면에표시	양호				양호	양호	양호	양호	합			
5	표시											합	

검사증명서 (Chứng chỉ kiểm tra của nhà máy)

수요자(Customer) : (주)드림에스티
증명서 번호(Certification No.) : DIQ-2406-0668
발행일자(Date of Issue) : 2024-06-14

Trụ sở : 111. Daesong-ro. Nam-gu.
Pohang-si. Gyeongsangbuk-do
Số / FAX : 054)285-7251~4/054)285-1250
Văn phòng / SEOUL 02)312-2201~4
BUSAN : 051)411-6561~2
TAEGU : 053)756-1201~5

제품번호		제품군	제품규격		치수	길이	중량	본수	제품번호	제강번호
Mã sản phẩm		Nhóm sản phẩm	Tiêu chuẩn		Kích thước	Chiều dài	Trọng lượng	Số lượng	Mã sản phẩm	Số mẻ luyện
단위 (Đơn vị)					mm	mm	kg	PCS		
VCG422-01		LA-ROUND BAR	AISI4140		32.00	6,680	9.070	213	VCG422-01	SF80427
합계 (Tổng cộng)							9.070	213		

제품번호		Thành phần hóa học													
Mã sản phẩm		C	Si	Mn	P	S	Ni	Cr	Mo	Cu					
Spec.	Max.	43	35	100	35	40		110	25						
	Min.	38	15	75	0	0	0	80	15	0					
단위 (Đơn vị)		X100	X100	X100	X1000	X1000	X100	X100	X100	X100					
VCG422-01		39.43	18	77.4	10.4	5	2.1	86.5	16.6						

- End Line -

DAECHANG CORP.

Jeongseong Kim CEO

Remark : WE HEREBY CERTIFY THAT THE MATERIAL HEREIN HAS BEEN MADE IN ACCORDANCE WITH
THE ORDER AND SPECIFICATION. NO REPAIR WELDING
ORIGIN : MADE IN KOREA

Hyun Jik Lee

QUALITY ASSURANCE TEAM MANAGER

검 사 증 명 서

Chúng nhận kiểm tra



(주) 대강선재

DAE KANG IRON WIRE CO.LTD.

고객명 / CUSTOMER : 신진화스너공업㈜

증명서번호 / CERTIFICATE NO. : 2025451070

trụ sở chính / Nhà máy : Gadal 1-ro, Quận Gangseo, Busan,

Hàn Quốc - Điện thoại: 82-51-831-215558 FAX :

82-51-831-2159 Sản xuất tại Hàn Quốc

Chúng tôi xin xác nhận rằng không có vật liệu nào từ Nga

được sử dụng

수주번호 / ORDER NO. :

발행일자 / DATE OF ISSUE : 2025.10.24.

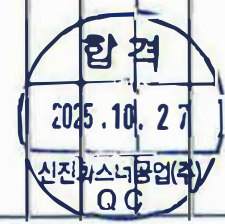
품명 / COMMODITY : AISI51820 HD 원형

규격 / SIZE : 19.50

HS CODE : 7229.90

치수 DIAMETER	제품 중량 WEIGHT	LOT NO.	SPEC	물 리 적 성 질 PHYSICAL PROPERTY								화 학 성 분 CHEMICAL COMPOSITION(%)											비고 REMARKS
				인장강도 T.S	항복강도 Y.S	연신율 E.L	단면수축율 R.A	경도 HARDNESS	구상화 S.A	탈탄 DECARBURIZATION	외관 APPEARANCE	C	SI	Mn	P	S	NI	Cr	TI	Cu	B	AL	
mm	kg			kgf/mm²		%		HRB		mm		*100	*100	*1000			*100					*10000	
19.50			MAX										35	90	35	40		90			30		
19.47			MIN	40				60				20	15	70			70			5			
19.48	1.968	SF93103	VDR1810151000	63.2		22.1	58.8		OK	17.4%	양	21	20	75	14	5		74			21		
19.48	1.956	SF94576	VDU8000181000	63.7		21.5	58.1		"	"	양	21	19	73	12	6		73			20		
19.48	1.965	SF94576	VDU8000191000	63.7		21.5	58.1		"	"	양	21	19	73	12	6		73			20		
19.48	1.955	SF94576	VDU8000211000	63.7		21.5	58.1		"	"	양	21	19	73	12	6		73			20		
19.48	1.966	SF95651	VDX6020041000	62.9		22.6	59.3		"	"	양	21	19	74	8	5		73			19		
19.48	1.970	SF95651	VDX6020071000	62.9		22.6	59.3		"	"	양	21	19	74	8	5		73			19		
19.48	1.982	SF95651	VDX6020081000	62.9		22.6	59.3		"	"	양	21	19	74	8	5		73			19		
TOTAL	13,762																						

DAECHANG CORP.



No Repair Welding

Manufacture Process :

Basic Oxygen

상기 鋼材는 규정에 따른 시험결과 합격하였음을 증명합니다.

WE HEREBY CERTIFY THAT THE MATERIAL DESCRIBED ABOVE HAS BEEN TESTED IN ACCORDANCE WITH THE SPECIFICATION.

CEO

Jung Joo yong

CHIEF OF QUALITY INSPECTION SECTION

주식회사 대강선재

검 사 증 명 서

Chứng nhận kiểm tra



(주) 대강선재

DAE KANG IRON WIRE CO.LTD.

고객명 / CUSTOMER : 신진화스너공업㈜

: 2025451170

본사/공장 : 부산광역시 강서구 가달1로7(생곡동)

수주번호 / ORDER NO.

증명서번호 / Số chứng nhận

Trụ sở chính / Nhà máy : Gadul 1-ro, Quận Gangseo, Busan,

품명 / COMMODITY : AISI51B20 HD 원형

발행일자 / Ngày cấp

: 2025.11.20.

Hàn Quốc - Điện thoại: 82-51-831-2155~8 FAX :

규격 / SIZE : 19.50

82-51-831-2159 Sản xuất tại Hàn Quốc

HS CODE : 7229.90

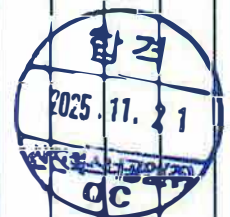
Chúng tôi xin xác nhận rằng không có vật liệu nào từ Nga được sử dụng

치수 DIAMETER	제품 중량 WEIGHT	물리적 성질 PHYSICAL PROPERTY										화학적 성분 CHEMICAL COMPOSITION(%)											비고 REMARKS
		LOT NO.	SPEC	인장강도 T.S	항복강도 Y.S	연신율 E.L	단면수축율 R.A	경도 HARDNESS	구상화 S.A	탈탄 DECARBURIZATION	외관 APPEARANCE	C	SI	Mn	P	S	NI	Cr	TI	Cu	B	AL	
mm	kg			kgf/mm²	%		HRB			mm	mm	*100	*100	*1000			*100					*10000	
19.50			MAX										35	90	35	40		90			30		
19.47			MIN	40			60					20	15	70			70			5			
19.48	1.177	SF95651	VDX6020131002	62.9		22.6	59.3		OK	19.48	양	21	19	74	8	5		73			19		
TOTAL	1.177																						

DAECHANG CORP.

Jeongseong Kim

CEO



Không có hàn sửa chữa, Quy trình sản xuất: Oxy cơ bản

상기 鋼材는 규정에 따른 시험결과 합격하였음을 증명합니다.

Chúng tôi xin chứng nhận rằng vật liệu được mô tả ở trên đã được kiểm tra theo đúng đặc điểm kỹ thuật.

Jung Joo yong

CHIEF OF QUALITY INSPECTION SECTION

주식회사대강선재

Số chứng nhận / 증명서번호 : 2025.12.05 -8

Ngày cấp / 발행일자 : 2025.12.05

Khách hàng / 수 요 가 : 신진화스틸공업(주)

Tên hàng / 품 명 : AISI 51B20 HD

19.50 mm

Tổng trọng lượng 총 중 량 : 1,870 Kg

Tiêu chuẩn / 규 격 : AISI

검 사 증 명 서

Chứng nhận kiểm tra

대성스틸(주) / DAE SUNG STEEL CO., LTD

Hwaseong Mado Cheongwon Bio valley #1249, Hwaseong-si,
Gyeonggi-do, Republic of Korea

TEL : 031-366-9723

FAX : 031-366-7144

Số nhà máy 소재번호		선경	중 량	길이	형상	Physical Properties / 물리적 성질 검사					Sự khử cacbon			Appearance
		DIA	Weight	LENGTH	Roundness	T.S	EL	R.A	S.A	Độ cứng	DMF	DMT		
Số lô nhiệt.	Số cuộn	mm	Kg	mm	mm	kgf/mm ²	%	%	Hạng	HRB	mm	mm		
rộng số kỹ thuật	rối đa	0.0			외관치수									
	tối thiểu	-0.03												
SF98284	VEB21400101	19.472	1,870		OK	19.48	65.0	17.0	63.5		84.1	NONE	NONE	GOOD
Total			1,870											

MILL NO. 소재번호		화합성분 (Chemical Composition)[%]												
Heat No.	Coil No.	C	Si	Mn	P	S	Cr	Mo	Ni	Cu	Sol-Al	B	Ti	
		X100			X1000		X100				X1000	X10000	X1000	
SPEC	MAX	22	30	90	35	40	90					30		
	MIN	17	15	70			70					5		
SF98284	VEB21400101	21.3	18.0	73.3	12.6	6.0	72.2					17		

* 상기 강재는 규정의 시험을 행하여 합격하였음을 증명합니다.

Chúng tôi xin chứng nhận rằng vật liệu đã được kiểm tra theo đúng quy định của đặc
điểm kỹ thuật Mã số Cn 6 chữ số: (Mã hải quan HS 7229.90).

Chúng tôi xin xác nhận rằng không có vật liệu nào từ Nga được sử dụng

Sản xuất tại Hàn Quốc

Jeongseong Kim

Trưởng phòng QC
(Kiểm soát chất lượng)

C.Y.LEE

DAECHANG CORP.



검 사 증 명 서

Chứng nhận kiểm tra



(주) 대강선재

DAE KANG IRON WIRE CO.,LTD.

고객명 / CUSTOMER : 신진화스너공업㈜

증명서번호 / CERTIFICATE NO. : 2025451110

본사/공장 : 부산광역시 강서구 가달1로7(생곡동)
Trụ sở chính / Nhà máy 7, Gadailro, Gangseo-gu, Busan, Korea

수주번호 / ORDER NO.

발행일자 / DATE OF ISSUE : 2025.11.05.

Điện thoại 82-51-831-2155-8 FAX 82-51-831-2159

품명 / COMMODITY : AISI51B20 HD 원형

Sản xuất tại Hàn Quốc

규격 / SIZE : 23.50

Chúng tôi xin xác nhận rằng không có vật liệu nào từ Nga được sử dụng.

HS CODE : 7229.90

치수 DIAMETER	제품 중량 WEIGHT			물리적 성질 PHYSICAL PROPERTY								화학적 성분 CHEMICAL COMPOSITION(%)												비고 REMARKS
				인장강 도 T.S	항복 강도 Y.S	연신율 E.L	단면 수축율 R.A	경도 HARD- NESS	구상화 S.A	탈탄 DECARBURIZA- TION	외관 APPEAR- ANCE	C	Si	Mn	P	S	Ni	Cr	Ti	Cu	B	AL		
실선경 SIZE			LOT NO.	SPEC																				
	mm	kg				kgf/mm ²	%	HRB						*100	*100	*1000		*100				*10000		
23.50				MAX											34	90	35	40		70			30	
23.47				MIN	43			60						20	15	70			70			5		
23.48	1.974	SF94576	V007970221000		58.7		32.5	5.7		OK	23.48	외관치수	21	19	73	12	6		73			20		
DAECHANG CORP. Jeongseong Kim CEO 합격 2025.11.06 신화산업(주) QC																								
TOTAL	1.974																							

DAECHANG CORP.

Jeongseong Kim CEO



No Repair Welding

Manufacture Process :

Basic Oxygen

상기 鋼材는 규정에 따른 시험결과 합격하였음을 증명합니다.

WE HEREBY CERTIFY THAT THE MATERIAL DESCRIBED ABOVE HAS BEEN TESTED IN ACCORDANCE WITH THE SPECIFICATION.

Jung Joo Yong

CHIEF OF QUALITY INSPECTION SECTION

주식회사대강선재

CEO